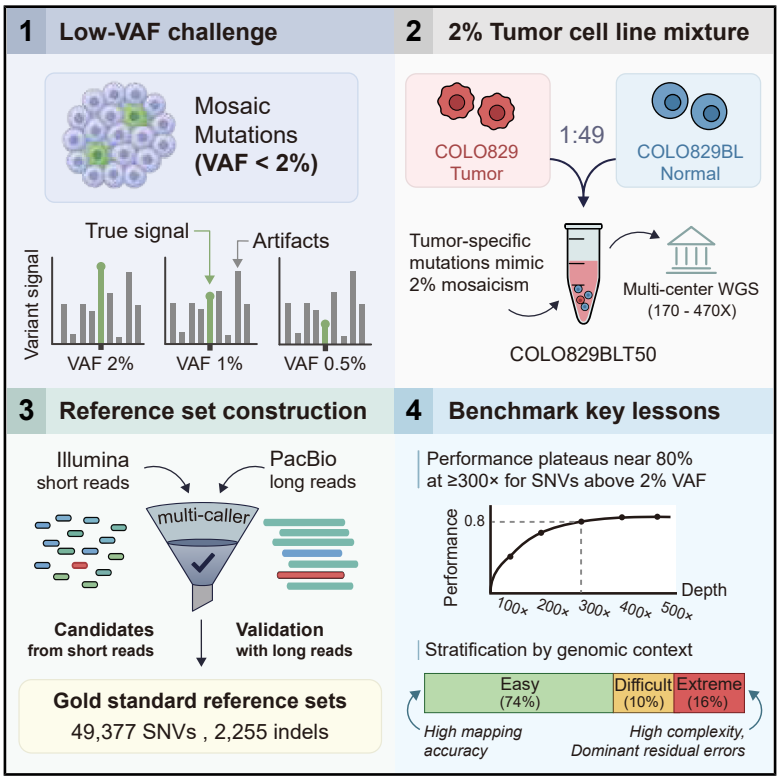
Cell Genomics

Resource

Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures

Graphical abstract



Highlights

- Multi-institution benchmark of low-VAF mosaic mutation detection by deep-coverage WGS
- Cell-line mixtures with Illumina/PacBio cross-validation define a robust truth set
- Sensitivity and precision plateau near 80% at ≥300× for SNVs above 2% VAF
- Reference variant sets and best practices provide a framework for mosaicism analysis

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In brief

Ha et al. benchmark detection strategies for mosaic mutations with low variant allele fractions (VAFs) using short- and long-read whole-genome sequencing. By cross-validating Illumina-based calls against PacBio HiFi data using cell-line mixtures generated across five institutions, they construct stratified reference sets, define realistic detection limits, and establish best practices for genome-wide analysis of low-VAF (<2%) variants in non-cancer tissues.

Ha et al., 2026, Cell Genomics 6, 101390
November 11, 2026 © 2026 The Authors. Published by Elsevier Inc.
https://doi.org/10.1016/j.xgen.2026.101390

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Please cite this article in press as: Ha et al., Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures, Cell Genomics (2026), https://doi.org/10.1016/j.xgen.2026.101390

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Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures

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https://doi.org/10.1016/j.xgen.2026.101390

SUMMARY

Mosaic mutations in normal tissues occur at low variant allele fractions (VAFs), complicating detection. To benchmark variant-calling strategies, the SMaHT Network created a cell-line mixture (1:49) and produced ultra-deep short- and long-read whole-genome sequencing data across five centers (170×–470× per center). We assembled a reference set of mosaic SNVs and indels, cross-validated with long-read data to expose short-read limitations. We also partitioned the genome by mappability to examine the impact of genomic context, added a negative reference set, and accounted for culture-derived mutations. When seven institutions applied twelve pipelines, callsets were largely discordant across tools and replicates, partly reflecting stochastic sampling of low-VAF mutations in biological replicates. For SNVs with VAFs above 2%, sensitivity and precision approached 80% at ≥300×, with little gain from additional sequencing. This work establishes a framework and best practices for reliably detecting low-VAF mutations in non-cancer tissues and provides a valuable resource for the community.

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Please cite this article in press as: Ha et al., Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures, Cell Genomics (2026), https://doi.org/10.1016/j.xgen.2026.101390

INTRODUCTION

Post-zygotic somatic mutations arise throughout life and are inherited by a subset of cells within an individual, leading to somatic mosaicism.[1] The clonality of somatic mutations varies widely, depending on the timing of acquisition, tissue lineage, and clonal expansion dynamics.[2] Standard bulk whole-genome sequencing (WGS) at ~30× coverage lacks the sensitivity to detect mutations with low variant allele fractions (VAFs) below 10%, where most mosaic variants fall into, necessitating higher depth of sequencing to capture both early clonal and recently acquired subclonal events. When accurately identified, these mosaic mutations can provide insights into human development, aging, and tissue-specific cellular dynamics. In addition, identification of mosaic variants in non-cancer samples is taking on increasing clinical significance, given the increasing number of pathogenic conditions caused by mosaic mutations in the heart,[3] liver,[4] and brain.[5]

However, detecting low-VAF mosaic variants—e.g., those below 5%—remains technically challenging[6,7] with validation rate remaining modest (~10%–40%). In the presence of sequencing errors and systematic artifacts, the signal-to-noise ratio worsens as the VAF decreases, making variant identification progressively more difficult. For cancer mutations, clonal expansion results in increased VAFs, and the availability of matched non-cancer controls enables the filtering of many technical artifacts (e.g., alignment errors shared between tumor and normal samples) as well as shared germline mutations. In contrast, identifying mosaic variants from a single sample without a matched control, as is frequently done in normal tissues, is substantially less accurate due to the inability to distinguish true variants from technical noise and germline mutations.[6,7] The challenge of identifying low-VAF mosaic variants is increasingly important in clinical diagnostics, given the growing number of non-neoplastic conditions such as vascular malformations and overgrowth disorders driven by these types of mutations.[8]

Comprehensive benchmarking of mosaic mutation detection remains limited, especially across the whole genome and those in extremely low VAFs (<5%). Prior studies using cell-line mixtures have yielded valuable insights but were largely confined to detecting relatively high VAF variants in limited genomic regions, e.g., exome or "high-confidence" regions that exclude repetitive or low-complexity sequences entirely.[7,9]

Here, we designed a comprehensive benchmark with the human melanoma cell line COLO829 in an extremely low ratio (tumor:normal ratio of 1:49) and generated a ground-truth reference set of mosaic single-nucleotide variants (mSNVs) and indels (mIndels). An accurate reference set is made possible by ultra-deep WGS generated by the Somatic Mosaicism across Human Tissues (SMaHT) Network, and ~500× of PacBio data allowed us to validate the mutations identified from Illumina data, such as those in "low-confidence" regions in which accurate read alignment is error prone. This orthogonal validation approach substantially increases the quality of the reference set, revealing limitations of Illumina-only benchmarking studies and enabling a comprehensive, genome-wide evaluation of detection methods.

Using the reference set with ultra-low VAFs, we analyze Illumina WGS data sequenced at multiple centers, with a combined coverage of >1,700×. We assessed the consistency of variant calls across several widely used algorithms and evaluated the performance of analysis pipelines from several independent groups within the SMaHT Network. We also discovered the presence of cell culture-induced mutations that are often overlooked in cell-line mixture studies. Furthermore, for additional insights on the factors that impact the detection of low-VAF variants, we analyzed a mixture of six well-characterized HapMap samples, for which known germline variants served as mosaic variants to be detected. The cell-line mixtures and validated reference sets are publicly available (https://data.smaht.org) to support the community in developing and benchmarking low-VAF variant detection approaches.

RESULTS

Construction of a high-confidence reference set using short- and long-read platforms for a cell-line mixture

We created COLO829BLT50, a biologically controlled cell-line mixture composed of 2% COLO829 (melanoma) and 98% COLO829BL (matched normal). This design preserved the original germline haplotypes while introducing COLO829-specific mutations at 1% or 2%, corresponding to heterozygous and homozygous COLO829-specific mutations, respectively (Figure 1A). Cytogenetics shows a complex aneuploid karyotype for COLO829 and a predominantly diploid karyotype for COLO829BL with a minor tetraploid subpopulation (STAR Methods and Figure S1). Because COLO829 is aneuploid, local copy-number changes and loss of heterozygosity (LOH) can shift locus-specific VAFs; accordingly, we base downstream analyses on empirically measured VAFs from high-coverage sequencing rather than diploid theoretical expectations. Then, COLO829BLT50 was independently sequenced at five genome characterization centers (GCCs), with calculated average depths ranging from 170× to 470× (Table S1; Figure S2), providing a robust and realistic resource for benchmarking low-VAF variant detection.

To construct the reference set, we first aimed to identify tumor-specific mutations present in COLO829 and absent in COLO829BL. We sequenced each cell line to high depth using Illumina short reads (COLO829: 370×; COLO829BL: 280×) and PacBio long reads (COLO829: 170×; COLO829BL: 330×), with PacBio depths a little lower given that they are used primarily for orthogonal validation rather than *de novo* discovery.

A series of quality-control measures were applied to the sequencing data, and key platform-specific features were systematically assessed (Figure S3). The reference mutation set generation proceeded in two stages: initial candidate detection using Illumina data followed by validation with PacBio long reads (Figure 1B).

To maximize sensitivity in initial discovery with short reads, we applied four complementary Illumina-based methods spanning diverse detection principles: Mutect2 and Strelka2, two widely used somatic variant callers (Mutect2 employs a Bayesian somatic genotyper that uses local assembly of haplotypes, whereas Strelka2 combines a probabilistic somatic model with adaptive indel error estimation and haplotype modeling); RUFUS, a k-mer-based detector free from reference bias; and VarNet, a deep learning-based image classifier.

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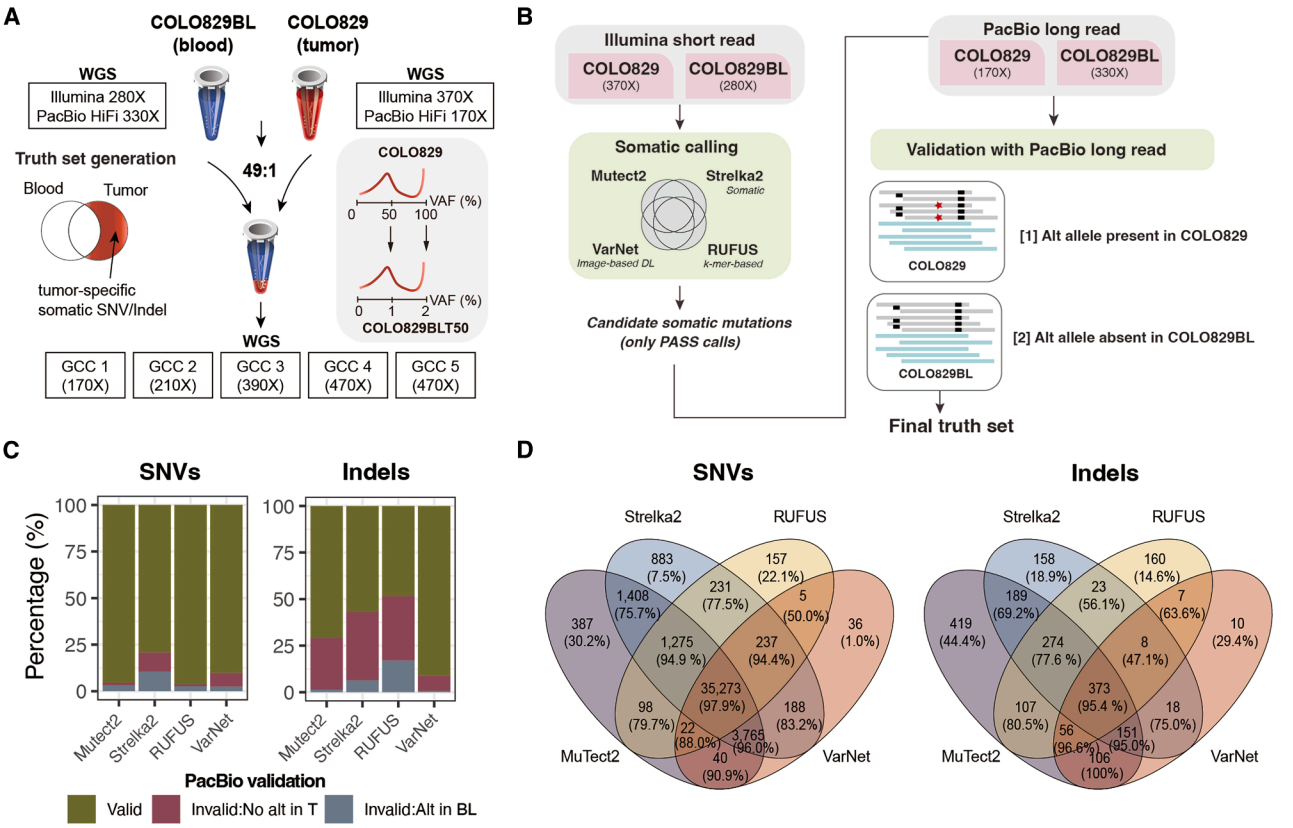


Figure 1. Generation of a high-confidence low-VAF somatic mutation reference set using a melanoma cell-line mixture

(A) COLO829 (tumor) and its matched normal COLO829BL (blood) were mixed in a 1:49 ratio to create COLO829BLT50, a reference material with mosaic mutations at expected VAFs of ~1%–2% and sequenced independently across five genome characterization centers (GCCs) at 170×–470× coverage.

(B) Reference set generation involves two steps: candidate variant identification from Illumina short-read data using four complementary somatic callers (Mutect2, Strelka2, RUFUS, and VarNet), followed by validation with PacBio long reads. A candidate variant was considered valid if the alternate allele was present in COLO829 and absent in COLO829BL.

(C) PacBio validation outcomes of SNV and indel candidates from each caller.

(D) Venn diagrams showing the overlap among validated SNV (left) and indel (right) calls across the four methods.

Variants identified by any of these methods were merged into a unified candidate set, comprising 61,649 SNVs and 4,475 indels.

Although increased sequencing depth improves the quality of variant calls, platform-specific errors persist,[10] especially in genomic regions in which read alignment is imperfect or copy number alterations cause a rare germline variant to appear as somatic. Thus, each candidate COLO829-specific mutation was subsequently evaluated for support in long-read data to eliminate short-read-specific artifacts and other systematic errors. A variant was considered validated if the alternate allele was detected in COLO829 and absent in COLO829BL in PacBio data. PacBio HiFi (CCS) reads have a low per-base error rate (~10^{-3}), comparable to Illumina. However, at the long-read depth used here (~500× combined across COLO829 and COLO829BL), the chance of observing at least one sequencing error matching the specific alt allele at any given position is high ($P(\geq 1 \text{ error}) = 1 - (1 - 10^{-3}/3)^{500} \approx 0.15$), making single-read "validation" overly permissive. Therefore, we required a minimum of two supporting PacBio reads to validate candidate mutations, which reduces its chance occurrence to ~0.01. In addition, because long-read sequencing can generate stutter-associated indel artifacts,[11] particularly near homopolymers and simple repeats, we employed COLO829BL Illumina data to remove variants lacking tumor specificity for indels.

Given the expected clonality of true tumor-specific mutations in the COLO829 cell line, most SNV candidates were consistently validated using PacBio long reads, confirming their authenticity where short-read-specific artifacts such as misalignment errors were eliminated. Validation rates for SNVs were high across methods (median of 92%; range: 79%–96%; Figure 1C; Table S2). The unvalidated calls comprised candidates lacking laternative allele support in PacBio COLO829 (4.3% in median) and those excluded due to the presence of the alternative allele in PacBio COLO829BL (median, 2.8%).

In contrast, only 63% of indel candidates were validated (median; range: 48%–91%), with most unconfirmed indels not supported by tumor long-read data (median: 30.5%). Conversely, a subset of indels including those identified by methods such as

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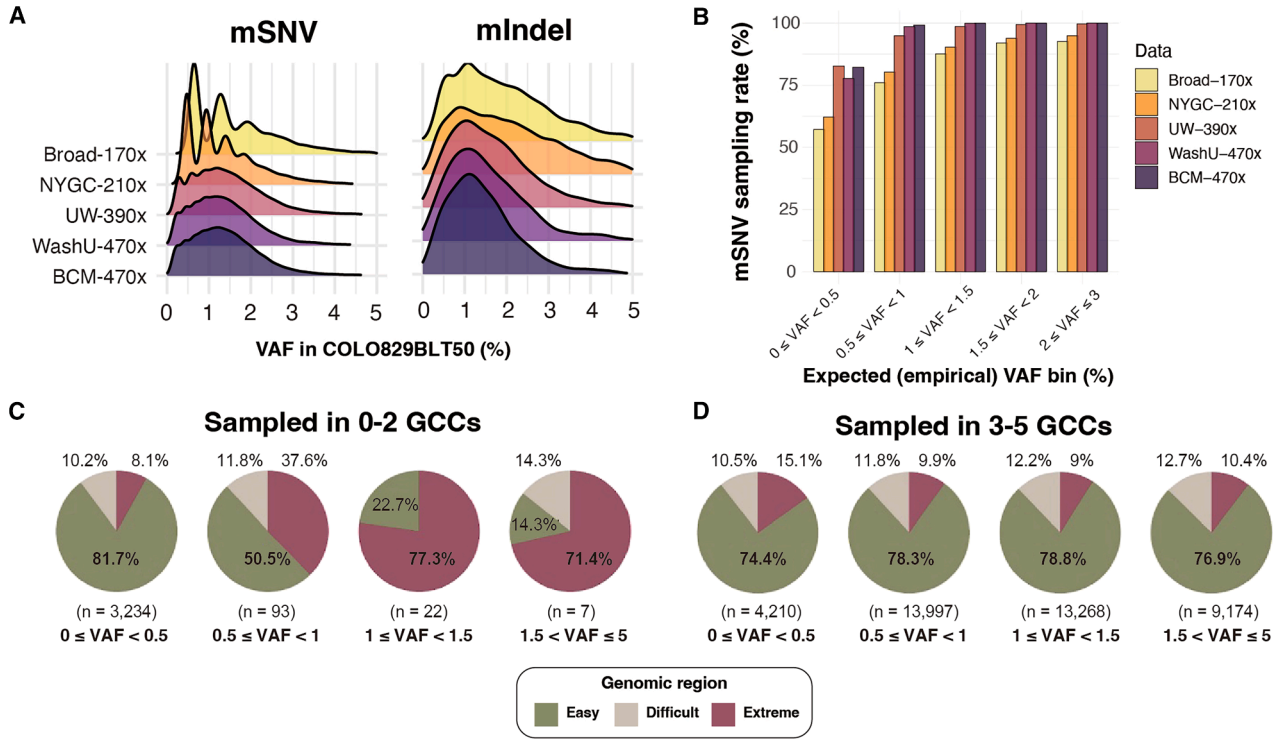


Figure 2. VAF distribution and sampling consistency of reference set variants across independent COLO829BLT50 datasets

(A) VAF distributions of reference set SNVs and indels as represented in five independently generated Illumina datasets from different sequencing centers (Broad, NYGC, UW-SCRI, WashU-VAI, and BCM). Each density plot reflects the observed VAF distribution of reference set variants captured in the respective dataset.

(B) For each dataset, sampling rate was computed as the fraction of truth-set mSNVs with non-zero observed VAF. Variants were binned by an empirical "expected VAF," defined as the mean VAF across the two highest-depth COLO829BLT50 datasets (WashU-470× and BCM-470×), which provides a depth-stabilized reference without assuming diploid copy-number states. Analyses were restricted to variants with empirical expected VAF >0.002 (above the minimum non-zero VAF representable by a single alternate read at ~500×).

(C) Variants sampled in ≤2 datasets.

(D) Variants sampled in ≥3 datasets.

Pie chart labels indicate the proportion of variants per region and the total variant count (*n*).

Strelka2 and RUFUS were also found in the matched normal (median: 6.4% and 16%, respectively), suggesting decreased specificity in some genomic contexts. These likely reflect misalignment issues or shared background signals in regions that are more difficult to resolve, highlighting the ongoing challenge of accurate somatic indel detection in such regions using short-read data alone—even in clonal cell-line models.

Validated SNVs and indels from all methods were then combined into the final reference set, comprising 44,005 SNVs, 961 insertions, and 1,098 deletions (Figure 1D). The vast majority of validated SNVs (over 96%) were identified by two or more callers out of four, reflecting strong consensus among tools for SNV detection, although some of the variants called by a single algorithm were also validated (e.g., 883 were found only by Strelka2). These validated SNVs detected by only one caller predominantly had a low VAF (<10%) in the tumor cell line, while SNVs with higher VAFs were robustly detected by three or four variant callers (Figure S4). Unlike SNVs, about 36% (747 of 2,059) of the validated indels were unique to a single method, underscoring the method-specific sensitivities and the inherent challenges of indel detection in short-read data. These findings highlight the importance of incorporating diverse variant-calling strategies and long-read data to comprehensively capture true somatic variants, particularly indels, across the genome, enabling both high sensitivity and precision, facilitating rigorous benchmarking of extremely low-VAF mosaic mutations in the COLO829BLT50 model.

Sampling variability for low-VAF mosaic mutation across independent datasets

To benchmark ultra-low-VAF variant detection performance, we utilized five independent Illumina WGS datasets of the COLO829BLT50 mixture generated at five GCCs, with depths ranging from 170× to 470× (Table S1). We confirmed that the reference SNVs and indels showed VAFs limited to extremely low levels, mostly under 2% (Figure 2A). For SNVs represented at lower depths (e.g., 170×), VAFs were more tightly concentrated around 1%–2%, yielding pronounced peaks that likely reflect limited allele support. With increasing coverage, these peaks were attenuated, and the distributions smoothed. Unlike SNVs, indel VAFs remained widely dispersed even in low-depth

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datasets, consistent with the increased mapping uncertainty of short reads containing insertions or deletions.

The characterization of low-VAF mosaic mutations in bulk sequencing data poses two major challenges: (1) the mutation must be "sampled" during sequencing such that the alternate allele is represented in the data and (2) the variant caller must correctly identify the signal and retain it through filtering artifacts. To evaluate the first challenge, allele sampling, we assessed how many reference SNVs and indels were actually observed in each of the five datasets generated from independent sequencing experiments. As expected, the low-VAF SNV sampling rate was highly dependent on the VAF of each variant (Figure 2B). At 170× depth, >82% of SNVs with VAF >1% were sampled, whereas datasets at 470× achieved near-complete (~99%) sampling. Variants with expected VAF < 0.5% are captured at ~80% in the 390× and 470× datasets and at ~60% in the 170×–210× datasets (Figure 2B), consistent with the expected depth dependence of allele sampling in the ultra-low VAF regime. These findings highlight that ultra-low-VAF subclonal mutations are intrinsically difficult to capture in bulk sequencing, as variant fragments may be missed even at very high coverage. Sequencing depth must, therefore, be optimized to balance sensitivity and cost, with biological replicates offering greater benefit than further increases in depth.

We next examined whether genomic context further shapes the sampling of low-VAF variants. Using a stratification framework that partitions the genome into "easy" (74%), "difficult" (10%), and "extreme" (16%) regions—defined from a short-read variant-calling performance perspective (see section "additional resources" in STAR Methods)[12]—we identified genomic context as a second major determinant of sampling consistency, alongside VAF. Variants absent or observed in only one or two datasets were disproportionately enriched in difficult and extreme regions, a bias that became more pronounced at higher VAFs (Figures 2C and 2D). By contrast, variants consistently observed across ≥3 datasets mirrored the expected genomic distribution (easy: 74%–79%; difficult: 10%–12%; extreme: 9%–15%). These results demonstrate that alignment uncertainty in low-complexity regions remains a critical barrier to the reliable detection of ultra-low VAF variants.

Generation of negative controls for accurate error estimation

We established a negative control set to avoid misclassifying true variants as false-positives in the benchmark (Figure 3A). Absent such a framework, undetected but potentially real mutations would confound benchmarking accuracy. For example, some loci carried alternative alleles in Illumina reads but fell below the thresholds required for variant calling, others were detected in Illumina but lacked PacBio support due to insufficient long-read depth, and still others represented subclonal variants in the pure cell lines that were missed during reference set construction. Failure to account for such cases would produce misleading performance estimates and unfairly penalize variant callers (Figure 3A middle).

The negative control set comprised reference homozygous sites, defined as positions without alternate allele evidence in both COLO829 and COLO829BL (STAR Methods). Variants shared between the two cell lines, which are likely germline variants and unlikely to represent mosaic events in the COLO829BLT50, were also excluded. Positions that did not fall into either the reference set or the negative control set were annotated as *masked* and excluded from benchmarking owing to ambiguity in classification.

We categorized mSNV and mIndel sites into three groups.

1. Reference set (positive control): mosaic variants detected using Illumina data and validated by PacBio data, present in COLO829 and absent in COLO829BL.
2. Negative control: positions lacking alternate alleles in both cell lines or consistent with germline variation.
3. Masked (ambiguous): sites not confidently assigned to either the reference or negative control sets, and thus excluded from evaluation.

Identification of cell-culture-derived mutations

Although cell lines are generally assumed to maintain stable genotypes across passages, immortalized cells can acquire *de novo* mutations over time, leading to subtle genomic divergence between harvests.[13] During assessment of putative false-positives in the evaluation, we observed that many variants absent from the reference set exhibited biological features consistent with genuine mutations rather than technical noise, appearing at reference homozygous sites in both pure COLO829 and COLO829BL lines. In addition, mutational signature analysis of these variants in COLO829BLT50 revealed a strong ROS-associated signal (COSMIC signatures SBS18/17b). These signatures, already associated with artifactual mutations in cell cultures, were absent from *in silico* mixtures (Figure S5), indicating that some apparent false-positives may instead reflect genuine culture-acquired mutations.

In preparing the COLO829BLT50 mixture, the normal cell line (COLO829BL) was harvested at four time points (H0–H3) with minimal passaging and was assumed to be genomically stable (Figure 3B). The H0 harvest was used to define the COLO829BL genome for reference set construction, whereas H1 through H3 were used for generating the mixture datasets. This design left open the possibility that subclonal mutations arising during COLO829BL passaging—present in H1-H3 but absent from H0—would represent true mutations in the COLO829BLT50 mixture.

To capture potentially subclonal mutations and ensure a robust benchmark, we examined mutation calls detected in five biological COLO829BLT50 replicates but absent from the COLO829/COLO829BL-defined reference set. Mutations were grouped by the number of sequencing replicates they were called in, as we hypothesized that those found in all five call sets were more likely to represent true-positive COLO829BLT50 admixture mutations than those only called, e.g., in a single replicate. To further assess these putative admixture sites, we cross-referenced each site with PacBio data for COLO829BLT50 (390× coverage) as well as PacBio data for an unrelated control (200× coverage), to eliminate any false signals produced by sequencing or systematic errors. Applying a minimum requirement of two supporting PacBio reads (Figure S6), we identified 11,408 admixture-only SNV mutations with long-read support (Figure S6B, top bars),

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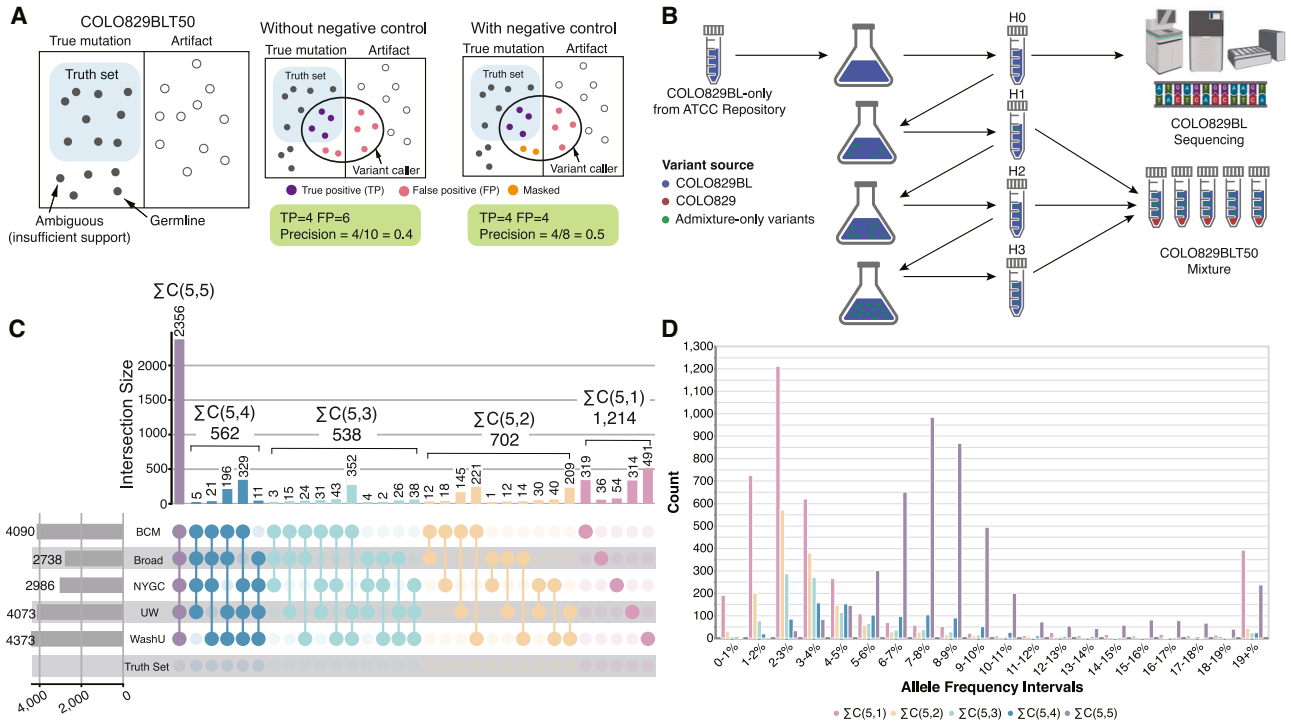


Figure 3. Importance of negative controls and evidence of cell-culture-derived mutations

(A) Conceptual schematic of the role of negative controls in variant benchmarking. Without a defined negative control set, true mutations that are not validated by PacBio data (possibly due to insufficient PacBio depth) may be incorrectly labeled as false-positives, which, in turn, artificially reduces precision. Incorporating both positive (i.e., the reference set) and negative controls enables more accurate classification by separating artifacts from ambiguous but potentially real mutations.

(B) COLO829BL harvest (H0) was sequenced prior to the subsequent cultures. The harvests, H1–H3, were used to create the COLO829BLT50 mixture, potentially allowing for emergence of somatic mutations in a normal lymphoblast cell line presumed to be stable. These mutations (visualized in green) were removed from the final reference set and masked from downstream analysis.

(C) The UpSet plot shows all possible intersections of validated, COLO829BLT50 admixture-specific SNV mutations found across five replicate datasets. Categories are color coded by the number of replicate datasets the SNV was found in; e.g., pink bars represent variants found only in a single replicate, while purple bars represent variants found in all five replicates.

(D) The histogram shows distributions of allele frequencies for the summed intersection sets grouped in (C), using the same color scheme. Strong unimodal distributions appear for variants appearing in fewer replicates (represented by pink, yellow, and green peaks), while a bimodal distribution appears for variants found in all five replicates (shown in purple), indicating the possibility of multiple subclonal expansions occurring between H0 and H3 harvests in COLO829BL. Indel versions of (C) and (D) can be found in Figure S8.

4.4% of which were erroneously supported (Figure S6B bottom bars). After removing the mutations also found in the unrelated control as well as those found in COLO829BL (see STAR Methods), we had a final count of 5,372 high-confidence admixture-only SNVs (Figure 3C). As expected, the largest number of these high-confidence mutations, 2,356, were observed in all five replicates, with 562, 538, 702, and 1,214 detected in four, three, two, or one replicate, respectively.

To assess indel mutations, we applied a similar PacBio read-support threshold, but additionally required that the alternative allele be the most frequently observed sequence at the site (major alternative allele), and cross-validated candidates against Illumina short-read tumor data (see STAR Methods). We found that the false signal rates in the control sample were much higher for indels (Figure S7, bottom bars), approaching almost 40% for a threshold of two supporting PacBio reads (Figure S7B). To minimize including false-positives, only indel mutations found in at least four of five sequencing replicate call sets were included in the final set, as validation rates for the control sample were observably lower than those in the COLO829BLT50 sample for those indels (Figure S7F). In total, 196 high-confidence indels were identified, of which 148 were called in all five sequencing replicates and 48 in four of five replicates (Figure S8). In examining the allele frequency distributions of our high-confidence admixture variants, we see a distinct mode for both SNVs (Figure 3D) and indels (Figure S8B) in the 7%–8% range, suggesting a subclonal expansion during passaging, rather than diffuse signal associated with technical artifacts. We experimentally validated three putative culture-acquired mutations consistently found across all five replicates using droplet digital PCR (ddPCR), which confirmed their increasing allele frequencies during COLO829BL passages and their presence in the final COLO829BLT50 mixture (Figure S9).

These sites were incorporated into the COLO829/COLO829BL reference set, annotated as culture-derived sites. Together, these analyses underscore that robust benchmarking cannot

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Table 1. Twelve single-sample-based mosaic mutation detection methods participated in the low-VAF mosaic mutation detection challenge

Method	Participant	Main algorithm	Main filtering method/note
M1	BCM	DRAGEN (v.4.3)	BAM processing: each BAM's header was edited to enable correct merging of data at sample level; variant calling: DRAGEN software v.4.3 was used to call variants with mosaic detection enabled; filtering: variants with PASS,MOSAIC, with AF $\leq$ 0.05 were retained, multi-allelic calls were excluded
M2	Broad	DRAGEN (v.4.3)	avoided RG issue (–use-single-read-group-for-bam-list); VAF <20% (–vc-mosaic-af-filter-threshold = 0.2); filtering: variants with PASS, MOSAIC were retained, multi-allelic calls were excluded
M3		DRAGEN-direct (v.4.3)	Same DRAGEN pipeline run directly on original CRAM
M4		GATK Mutect2 (v.4.1.6.0)	filtered against a panel of normals from 1000 Genomes and gnomAD; excluded regions with very low (<50×) or abnormally high (>1,000×) sequencing coverage, removal of residual germline variants using a one-sided exact binomial test assuming an expected germline VAF of $\geq$0.5, followed by a Benjamini-Hochberg correction with a q value cutoff of 10^{-5}[19]
M5	NYGC	Lancet2 (2.8.5-main-cfbc7d5143)—single sample mode	filtered against germline using HaplotypeCaller + GenotypeGVCFs; removed calls in gnomAD v.3.1.2 (>0.00001% AF) and High Coverage 1000 Genomes callset; excluded UCSC SimpleRepeats and SegDup regions; min and max depth thresholds = 0.65–1.25 times the mean sample coverage, allele depth $\geq$4, VAF $\leq$ 10%, strand, allele quality, mapping quality considered
M6	WashU-VAI	Mutect2—tumor only mode (GATK 4.4.0.0)—VAF	filtered against publicly available panel of normal and gnomAD; FilterMutectCalls; VAF < 2%
M7		Mutect2—tumor only mode (GATK 4.4.0.0)—dbSNP	filtered against publicly available panel of normal and gnomAD; FilterMutectCalls; removed calls in dbSNP v.151; remove calls with depth <50× or >1,000×; remove variants not located on canonical chromosomes
M8	UW-SCRI	Mutect2—tumor only mode	used the GATK panel of normals and gnomAD in the calling step to filter potential germline variants; FilterMutectCalls; PASS variants only filtered for VAF less than 40% with depth >20×
M9	Utah-TTD	RUFUS—lenient	First of 3 filter levels to reduce noise: (1) lenient; RUFUS only works with a matched normal sample (280× COLO829BL was used for control)
M10		RUFUS—moderate	(2) moderate; RUFUS only works with a matched normal sample (280× COLO829BL was used for control)
M11		RUFUS—strict	(3) strict; RUFUS only works with a matched normal sample (280× COLO829BL was used for control)
M12	DAC	MosaicForecast	removed calls in UCSC SimpleRepeats, SegmentalDuplication, MosaicForecast clustered regions

rely on pure cell lines alone. Whereas most prior cell-line-based benchmarks overlooked culture artifacts and ambiguous sites, our framework integrates curated positive and negative controls and explicitly validates uncertain loci. Such rigor is essential for ensuring that benchmarking truly reflects caller performance rather than hidden biases of the model system.

The SMaHT low-VAF (<2%) mosaic mutation detection challenge

To assess the effectiveness of current methods for low-VAF mosaic SNV and indel detection, we conducted a network-wide benchmarking challenge. This effort focused on testing somatic variant detection pipelines applied to single-sample Illumina WGS data from the COLO829BLT50 mixture, which contained mosaic mutations with VAFs below 2%.

Five biological replicates of the mixture, sequenced at depths ranging from 170× to 470×, were used to represent real-world whole-genome data with biological and technical variability (Table S1). In addition, five *in silico* COLO829BLT50 datasets were generated by mixing BAM files of the parental cell lines at defined total depths from 100× to 500×, enabling a controlled evaluation of depth-dependent performance. Because our analyses were limited to datasets with depths up to 500×, we restricted the truth set to variants theoretically observable at this depth, retaining 39,389 of 49,377 SNVs (pure cell-line-derived; 44,005 and cell-culture-derived: 5,372) and 2,059 of 2,255 indels (pure cell-line-derived: 2,059 and cell-culture-derived: 196) from the COLO829 for benchmarking (Table S3), including both pure cell-line- and culture-derived mutations. The excluded ultra-low-VAF subclonal mutations remain valuable for future studies targeting much higher depths, such as

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ultra-deep panel or duplex sequencing.[14] Given the mostly diploid karyotype of COLO829BL and its dominant contribution to the mixture (98%), we confirmed that the potential impact of aneuploidy in the cancer cell line COLO829 is effectively mitigated in the mixture (STAR Methods, Figure S1; Table S4).

A total of seven institutions participated, contributing twelve variant-calling approaches for both mSNVs and mIndels (STAR Methods). These pipelines included tools developed specifically for mosaic variant detection or adapted from frameworks originally developed for somatic mutation detection in cancer (Table 1; Table S5): MosaicForecast,[15] a machine learning model trained on read-level features, was included based on its previously reported high performance in exome-based benchmarks; Mutect2,[16] a widely used somatic variant caller, was run in tumor-only mode under three different configurations, varying post-calling filters based on VAF thresholds and depth cutoffs; Lancet,[17] a local-assembly-based method, was also tested for its ability to resolve complex or low-VAF mutations; and RUFUS,[18] a reference-agnostic sequence comparison tool, was evaluated for its alignment-free detection strategy. Although the challenge focused on single-sample variant calling, the limited use of the COLO829BL control sample was permitted for RUFUS. This allowance enabled evaluation of its sequence-based comparison algorithm, which bypasses alignment and may offer advantages in detecting mosaic variants in complex genomic regions.

Performance evaluation of the detection challenge

Systematic evaluation across sequencing depths is particularly critical for low-VAF variants (<5%), where the number of supporting reads is inherently small (theoretically only 1 at 100× or 5 at 500×). In practice, such signals are further obscured by sequencing inconsistencies and low-quality features, such as short read lengths, base-calling errors, and reduced base quality scores that exacerbate detection.

In the COLO829BLT50 mixtures, sensitivity of mSNVs was uniformly poor at 170× data, with only a few pipelines reaching ~50% in the 2%–3% VAF bin (Figure 4A). At 390× and higher depths, nearly half of the methods exceeded 70% sensitivity, with MosaicForecast (M12), Broad (M4) and UW-SCRI Mutect2 (M8), and RUFUS with matched control (M9–M11) performing best. Pipelines enforcing hard VAF cutoffs, such as WashU-VAI Mutect2 (M6) and BCM DRAGEN (M1) with strict VAF filters (<0.02 and <0.05, respectively), showed marked limitations. Because VAF is inherently depth dependent and shaped by dataset-specific noise, such fixed thresholds often suppressed true-positives. Precision at lower depth (170×) was poor in even 2%–3% VAF, where most of the methods could recover high precision at VAF >1% from 390× (Figure S10).

Among all evaluated tools, MosaicForecast provided the most balanced performance in terms of sensitivity and precision up to 210× (Figure 4B). At 470×, likelihood-based methods such as Mutect2 gained greater statistical support for true alleles and outperformed other approaches. These depth-dependent trends motivate the practical recommendations summarized in Box 1.

To further assess the effect of sequencing depth, we applied the same pipelines to five *in silico* COLO829BLT50 mixtures generated by combining pure cell-line BAM files at defined depths (100×–500×) (STAR Methods). Similar patterns were observed: sensitivity for variants with VAF <1% remained poor and did not improve significantly with higher depth. Gains in F1 score, reflecting the balance of sensitivity and precision, were evident up to 300× for VAF > 1%. For example, the F1 score of MosaicForecast (M12) increased from 0.15 at 100× to 0.48, 0.68, and 0.75 at 200×, 300×, and 400×, respectively, but plateaued at 0.74 at 500× (Figure S11; Table S6). Overall, F1 scores improved steadily with increasing depth (Table S7), although sensitivity remained the principal bottleneck, particularly for ultra-low VAF (<1%) variants across all pipelines.

For ultra-low-VAF mIndels, sensitivity remained low even at 470×, reaching only ~15% with RUFUS when applied with a matched control (Figure 4C; Figures S12 and S13). Single-sample approaches achieved ~10% sensitivity but with higher precision, reflecting a trade-off between recall and specificity. Given the biological importance of mIndels at low VAFs and the inherent mapping challenges of short reads spanning insertions or deletions, improved methods specifically designed to overcome this limitation are urgently needed.

We next evaluated a more practical scenario in which five independent groups (BCM, Broad, NYGC, UW-SCRI, and WashU-VAI) applied their own pipelines to independently generated COLO829BLT50 datasets. This design mimics real-world mosaic variant calling, enabling assessment of how variant profiles converge or diverge when the same biological material is analyzed across institutions for ultra-low-VAF mosaic mutations. Performance largely tracked with sequencing depth (Figure 4D), although some relatively lower-depth datasets, such as the 390× dataset from UW-SCRI, achieved F1 scores comparable to higher-depth datasets, underscoring the importance of both data quality and analysis strategy. Detected mutations were also highly distinct across groups, with thousands of true-positives unique to individual pipelines (Figure 4E). False-positives varied substantially, with little overlap across methods. These results suggest that intersecting multiple callers may reduce false-positives but at the expense of discarding many true variants, highlighting the need for more sophisticated algorithms for ultra-low-VAF mutation detection.

Low-VAF mutation detection in different genomic regions

Last, we examined how genomic context influences the detection of low-VAF mutations by using high-confidence variant sets comprising 38,680, 5,754, and 4,943 SNVs and 752, 699, and 804 indels in easy, difficult, and extreme regions, respectively (Table S3). To first assess whether VAF estimates differ between sequencing platforms, we compared Illumina and PacBio data across regions. VAFs for mSNVs were highly concordant in easy ($R = 0.99$, $p < 10^{-15}$) and difficult ($R = 0.98$, $p < 10^{-15}$) regions, but correlation was reduced in extreme regions ($R = 0.92$, $p < 10^{-15}$) (Figure 5A).

Indels showed lower overall concordance, with correlations of 0.97, 0.84, and 0.71 in easy, difficult, and extreme regions, respectively. These results indicate that even in highly clonal COLO829 samples where mutations typically have VAFs near 50%–100%, alignment ambiguity persists in short-read WGS. Such effects are expected to be exacerbated in mosaic contexts

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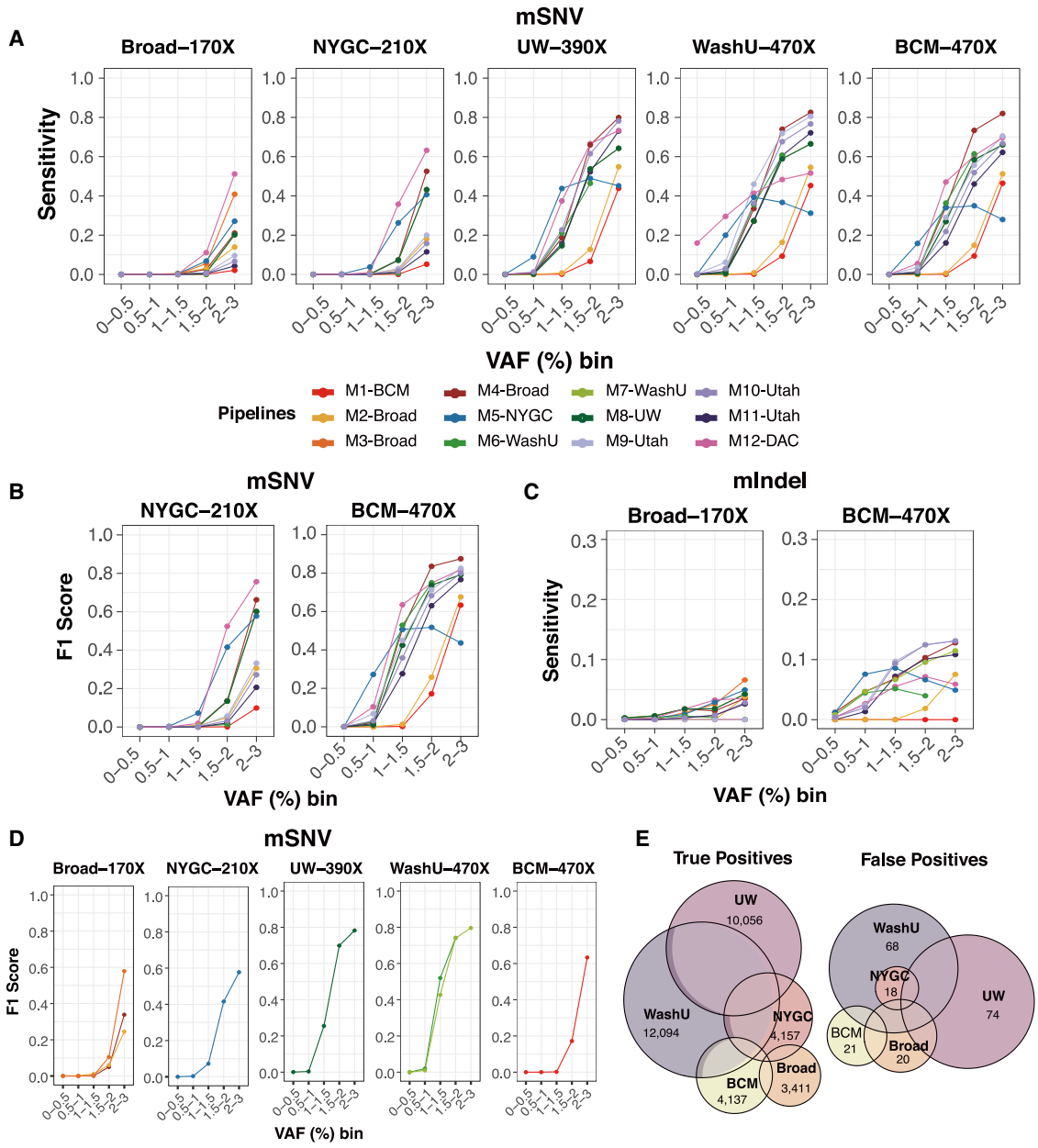


Figure 4. Performance benchmarking of low-VAF mosaic variant detection

(A) Sensitivity across VAF bins for 12 detection pipelines applied to five biological COLO829BLT50 datasets (170×–470×). The underlying performance values for all pipelines are provided in Table S6. All pipelines were single sample, except RUFUS, which was run with a matched control.

(B) F1 scores of mSNV detection in 210× and 470× are shown.

(C) Sensitivity of mIndels across VAF bins for two representative depths (Broad 170× and BCM 470×).

(D) F1 scores stratified by VAF bin across five independently generated datasets, each analyzed using a distinct institutional pipeline, reflecting the combined effects of data quality and detection method on overall performance.

(E) Overlap of true-positives (left) and false-positives (right) detected from the best five methods in (D) (M2 for Broad and M7 for WashU), with variant calls having VAF < 0.03. M6-WashU was not evaluated for VAF > 2% due to its filtration criteria.

such as COLO829BLT50, where low-VAF mutations are intrinsically limited by lower signal-to-noise ratios, further complicating accurate detection.

We next evaluated detection performance across genomic contexts. For mSNVs in easy regions, 1% VAF mutations achieved ~60% sensitivity and >90% precision, with F1 scores exceeding 0.9 for variants ≥2% VAF (Figure 5B; Figure S14), consistent with prior exome-based benchmarks.[7] In contrast, sensitivity declined sharply in difficult and extreme regions: at 1% VAF, F1 scores dropped to ~0.4, roughly half of those

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Box 1. Conclusions and best practices for genome-wide detection of ultra-low (<2%) VAF mosaic mutations

Key message

- Accuracy of genome-wide detection of mosaic mutations below ~2% VAF is influenced by allele sampling (sequencing depth), caller model assumptions, and genomic context. Best practices should, therefore, be depth aware, pipeline specific, and region stratified.
- Comparison of performance metrics for mosaic mutation detection across studies requires careful interpretation, as results depend on multiple parameters. These include whether analyses are restricted to "easy" regions or span the whole genome, how the reference set is defined (e.g., using a negative control set and accounting for culture-derived mutations), how sampling variability is incorporated, and whether VAF is specified by study design or estimated from pileup data.

Data design and sequencing depth

- There is diminishing return for higher depth, especially past ~300×, for SNVs with VAFs 1%–2% or below (Figure 6D).
- For SNVs with VAF ~2%, the best algorithms give 80% sensitivity at 300× (but only 40% sensitivity for 100×) (Figure 6D; Figure S11).
- For studies targeting ~1% VAF SNVs, ~400× WGS captures at least one supporting read for >80% of variants in the mixture benchmark (Figure 2B).
- Ultra-low VAF variants (<0.5%) remain poorly detected even at very high depth (≥500×) due to sampling limit and noise, not just insufficient coverage (Figures 2B and 4A).
- Biological or technical replicates rather than simply increasing depth may be more cost effective, as sampling variability limits detection and replicates can improve variant recovery more effectively (Figure 4D).

Analytical method selection based on sequencing depth and desired sensitivity/specificity

- At ≤300×, MosaicForecast provides the most balanced sensitivity-precision trade-off, reflecting the value of read-level feature modeling when evidence is limited (Figure 4A; Figure S11).
- At >300×, likelihood-based callers (e.g., Mutect2) with relaxed thresholds benefit most from added depth, gaining statistical support for true low-support alleles relative to artifacts (Figure 4A; Figure S11).

Genomic context and practical artifact control

- Performance varies sharply across easy, difficult, and extreme regions. We recommend reporting and optimizing pipelines separately by region, as complex regions dominate residual errors and depth requirements (Figure 5B).
- Reliable analysis in difficult and extreme regions requires explicit control of recurrent, alignment-driven artifacts, which can be identified across independent datasets and filtered to reduce false-positives while preserving true-positives (we provide one such filter, Figure 5D).

Indels require specialized strategies

- Ultra-low-VAF indel detection remains substantially more challenging than SNVs (single-sample sensitivity <0.15 in VAF < 2% range) (Figure 4C).
- Sensitivity improves with local-assembly approaches (e.g., Lancet) and/or reference-free k-mer evidence with matched controls (RUFUS), though performance remains severely limited in complex sequence contexts (Figure S14B).

observed in easy regions. Methods generally clustered into two categories: those favoring precision (e.g., DRAGEN [M2 and M3], MosaicForecast [M12]) and those favoring sensitivity (e.g., Mutect2-based pipelines [M6–M8]). Interestingly, single-sample methods (MosaicForecast [M12] and Mutect2 [M8]) outperformed matched-control approaches (M9–M11) in easy regions, suggesting that robust statistical or machine-learning models can distinguish true low-VAF mutations from artifacts without external controls. As all pipelines were evaluated using the same parameter settings regardless of genomic context, the observed performance differences highlight the potential value of region-specific filtering strategies to better address distinct error profiles across genomic regions.

In contrast, mIndel detection remained substantially more challenging (Figure 5C). Alignment-free or alignment-modified methods yielded the best performance, with sensitivity in easy regions reaching only ~0.3 at 1% VAF and rising to ~0.6 at 3% when matched controls were used (M9–M11). Lancet (M5), a local assembly method, performed best at intermediate VAFs (0.5%–1.5%), offering high precision with modest sensitivity. These findings underscore the urgent need for improved methods to address the alignment ambiguity of short reads at indel sites, as well as persistent sensitivity gaps for mSNVs in difficult and extreme regions. Overall, low-VAF mutation detection was constrained by artifacts that varied in impact across genomic contexts. By using long-read-supported reference sets, we were able to extend evaluation beyond easy regions and characterize the variant-calling performance in difficult and extreme regions, where context-specific artifacts shaped sensitivity and precision in distinct ways. These patterns provide valuable clues for developing more advanced methods for detecting ultra-low-VAF mosaic mutations across the whole genome.

Finally, because false-positives are disproportionately enriched in difficult and extreme regions, we developed a practical artifact-filtering workflow that identifies systematic, alignment-driven errors by checking whether the same candidate signals recur across independent datasets. Applying this cross-sample filter substantially reduced mSNV false-positives in difficult and extreme regions (50.3% and 59.4%, respectively), with minimal loss of true-positives (0.3% and 0.1%, respectively). Notably, this cross-sample artifact filter nearly eliminated spurious ultra-low-VAF (<0.5%)

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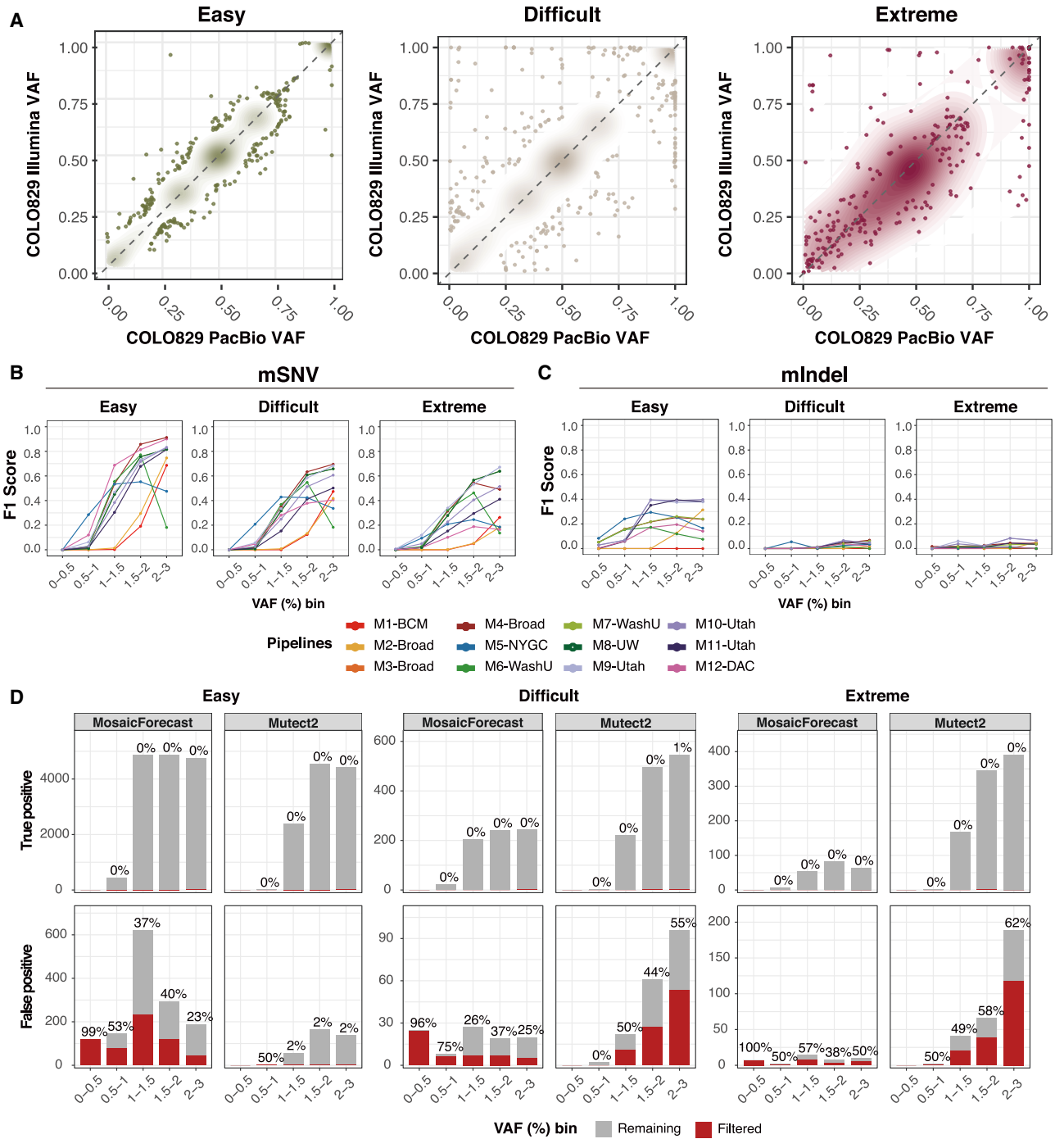


Figure 5. Impact of genomic region complexity on low-VAF mutation detection performance

(A) VAF comparison between PacBio and Illumina for COLO829 across three genomic regions. Two-dimensional density indicates the concentration of mosaic SNVs, with low-density outlier variants overlaid as points. Colors indicate genomic region category: easy, difficult, or extreme.

(B and C) F1 scores stratified by different genomic contexts (easy, difficult, and extreme). All pipelines were single sample, except RUFUS, which was run with a matched control. 470× BCM-generated COLO829BLT50 was used with ten detection pipelines. (B) mSNV and (C) mIndels.

(D) Stacked bar plots show true-positive (TP) (top) and false-positive (FP) (bottom) counts across VAF bins for easy, difficult, and extreme regions, comparing original callsets (total height) to the filtered callsets.

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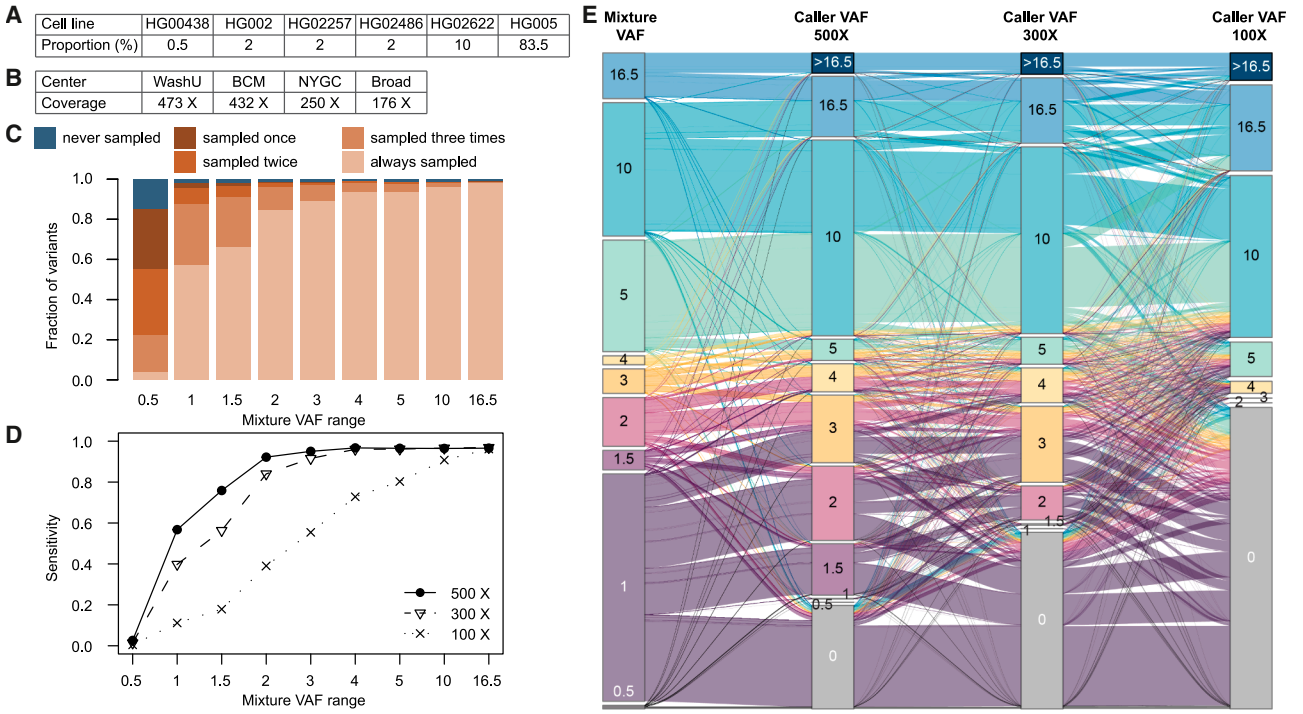


Figure 6. HapMap mixture analysis on sampling bias, sequencing depth, and VAF dynamics

(A) Proportion (%) of cell lines in the HapMap mixture sample.

(B) Average sequencing depth of genome-wide coverage of Illumina short-read data generated from technical replicates of the HapMap mixture at different centers.

(C) Sampling of somatic variants depends on their VAF. While >85% of variants with VAF $\leq$ 0.5 and >98% of variants between 0.5% and 1% were sampled at least once in the 4 short-read datasets, only 4% and 57% of variants were consistently sampled in all 4 datasets, respectively. With higher VAFs, variants are more reliably sampled in all datasets.

(D) Sensitivity of variant calling depends on the sequencing depth, with an increasing number of low-VAF variants called with deeper sequencing of 500× compared to 300× and 100× approaches. Subsampled 300× datasets showed comparable sensitivity to datasets aggregated to 300× from multiple 100× datasets subsampled from different datasets.

(E) Comparison between expected "mixture VAF" based on the proportion of variants in the HapMap mixture and "caller VAFs" calculated by a variant caller, i.e., Mutect2, across different sequencing depths. Variants are colored by their mixture VAF range. At lower sequencing depths, the caller VAF deviates further from the mixture VAF, and false-negative calls increase (indicated by gray bars labeled with 0).

signals while preserving true-positive counts across VAF bins (Figure 5D), providing a practical strategy to improve precision in settings—such as clinical pipelines—where false-positives are particularly costly. We provide this workflow as an easy-to-use, reproducible module for end users[20] (STAR Methods).

Impact of sequencing depth and sampling variability based on HapMap cell-line mixture

To further validate the consistency of analyses with COLO829BLT50 and to extend evaluation of detection limits beyond the 2% design of COLO829BLT50 up to ~15% VAF, we employed a HapMap-based mixture model comprising ~4.8 million reference SNVs. Six well-characterized HapMap cell lines (HG002, HG005, HG00438, HG02257, HG02486, and HG02622) were mixed to create target VAFs ranging from 0.25% to 16.5% (Figure 6A). Although the six sample-mixture design does not permit haplotype phasing, it enables the evaluation of a larger number of variants across a broad range of VAFs (Figures S15A and S15B). The reference set for the HapMap mixture benchmarking experiment was created from curated germline SNV calls from Genome in a Bottle and the Human Pangenome Reference Consortium (STAR Methods and Table S8). We generated a reference SNV set comprising ~4.8M SNVs with about a half (2.1M) at VAF $\leq$1%, with multi-allelic sites removed and restricted to high-confidence regions. HapMap mixture samples were sequenced up to ~470× short-read and ~100× long-read whole-genome coverage across four centers each (Figure 6B; Table S1).

For variants with expected VAFs >1%, over 98.8% were observed at least once out of four short-read datasets of the HapMap mixture and 92% could be consistently observed across all four replicates, reinforcing that even 250× coverage provides robust sensitivity (Figure 6C). By contrast, only 4% of variants with up to 0.5% expected VAF were consistently sampled across all samples, but 85% were identified at least once, predominantly by the >400× samples.

Building on these sampling rates, we evaluated variant calling with Mutect2[16] to assess how coverage and allele support translate into final mutation calls (Figure S15C). Notably, sensitivity was improved substantially between 100× and 300× from

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11% to 40% for 1% VAF variants and 39% to 84% for 2% VAF variants, but the gain is minimal beyond 300× (Figure 6D). This quantification of diminishing returns in variant detection sensitivity beyond 300× informed the SMaHT Network's decision to adopt 300× as the target depth for all bulk sequencing. To evaluate the impact of merging sequencing datasets across centers, we combined three independent 100× subsampled datasets from different centers to achieve an aggregate 300× coverage dataset and compared variant-calling performance with subsampled 300× datasets from individual datasets. Sensitivity was nearly identical to that of single-center 300× datasets (Figure S15C), supporting the feasibility of merging data across centers to achieve higher coverage without loss of performance.

Because the mixture proportions were precisely defined, we were also able to compare VAF expectations with VAFs observed and reported by somatic variant callers at different sequencing depths (Figure 6E; Figure S15B). In general, the VAF reported from Mutect2 (i.e., caller-VAF) was consistently higher than expected, reflecting sampling bias: variants that happen to receive more read support are more likely to be called. Overall, for 23% of variants, the caller VAF remained consistently within the expected mixture VAF range. However, most of these were high-VAF variants; only 4.7% of variants with expected VAF $\leq$5% were concordant, indicating that VAFs were often inflated, particularly at low frequencies. At 100× coverage, only 38 variants were called at the lowest reported VAF of 2%, with expected values between 0.5% and 4%, highlighting the limited precision of VAF determination for low-VAF somatic variants. This bias underscores the need for caution when interpreting VAFs from variant callers in applications where accurate quantification is critical.

DISCUSSION

Comprehensively detecting mosaic mutations remains one of the most difficult challenges in genome analysis, in part because it requires pushing the limit on distinguishing low-frequency mutational signals from the inherent background noise of high-throughput sequencing data. Although the cost of WGS has dropped significantly in recent years, simply increasing sequencing depth to improve the signal-to-noise ratio is not a cost-effective strategy. For example, in our benchmark, median sensitivity across methods for detecting mSNVs at 1%–2% VAF was approximately 0.5 even at ~500× coverage and declined sharply at lower VAFs (Figures S10 and S11). This highlights a clear point of diminishing returns, where further increases in depth yield only marginal gains and fail to capture mosaic mutations beyond those arising in the earliest cell divisions.

We generated an exceptionally high-coverage whole-genome dataset to assess the impact of sequencing depth, variability in replicate experiments, and algorithmic performance on mosaic variant detection. While WES[21] and targeted panels[22,23] can reach much lower VAFs at selected loci through ultra-deep sequencing, deep WGS uniquely enables genome-wide evaluation—including complex and repetitive regions—thereby supporting the callability- and context-aware benchmarking framework. Despite applying four Illumina-based methods for candidate identification, a small number of variants were inevitably missed due to imperfect sensitivity of the current algorithms. To address this, we introduced a negative reference set and incorporated carefully characterized cell-culture-derived mutations, enabling a more robust evaluation framework of low-VAF mosaic mutations—features largely overlooked in previous cell-line-based benchmarking designs.

Our results revealed that many Illumina-derived variants were not validated by PacBio reads, identifying important sources of false-positives in Illumina-only pipelines. Some false-positive calls were, in fact, rare germline variants in copy number-amplified regions. For example, in a region where the variant allele remains at one out of two copies while the reference allele is amplified by two additional copies, the germline variant would appear at VAF $\approx 1/(2+2) \approx 0.25$ if interpreted under a diploid assumption. Additional false-positives arose, for instance, in segmental duplication or repetitive regions, where alignment ambiguities cause reads from multiple similar regions to collapse onto a single locus, making mutations in one copy appear as low-VAF mosaic events.

Notably, we generated high-coverage PacBio HiFi long-read WGS data, enabling confident orthogonal validation of candidate variants, including those in repetitive or difficult-to-align regions. This dual-platform approach significantly improved the quality of our curated reference set and revealed key limitations of benchmarks based solely on short-read data, demonstrating that artifacts impact sensitivity and precision differently across genomic contexts. Future methods will need to explicitly model and filter such region-specific artifacts, and the benchmarking resources provided here establish a foundation for developing and validating next-generation mosaic variant callers. To facilitate such development, we provide a practical artifact filtering workflow (Figure 5D) together with the benchmark resources, enabling end users to systematically reduce recurrent false-positives in complex genomic regions.[12]

Recent GIAB/GA4GH/NIST benchmarking efforts have begun to address mosaic variants, primarily at VAFs $\geq$ 5%, and our study complements these resources by extending systematic benchmarking into the ultra-low-VAF regime (<2%) using analogous principles of validation, stratification, and reproducibility. Although this study focuses on mosaic SNVs and indels—the most prevalent forms of postzygotic variation—other mosaic variant classes, including CNVs, structural variants, and repeat expansions, are also clinically important and can have larger phenotypic effects. Many of the technical challenges highlighted here, such as copy-number-induced artifacts, alignment uncertainty, and context-dependent detectability, are likely to similarly impact accurate calling of these variant types, and the resources and principles developed in this study provide a foundation for extending benchmarking to them.

Although PacBio data were essential in building our reference set, obtaining high-depth PacBio HiFi data from tissues can be challenging in practice due to the requirement for high-molecular-weight DNA, which cannot be reliably extracted from many tissue types. However, our cross-platform validation has informed the development of more effective Illumina-only detection strategies, improving both accuracy and feasibility in diverse biological samples.

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To overcome the limitation of high-depth WGS, complementary technologies are essential. Single-cell WGS, for instance, has been used to identify rare mutations,[24,25] which can then inform the design of targeted panels for ultra-deep sequencing (e.g., >100,000×) across multiple tissues. Duplex sequencing, which enhances specificity by capturing both strands of a DNA molecule, offers another powerful strategy for detecting rare mosaic variants at the single-molecule level.[14] The disadvantage of duplex sequencing, however, is that one cannot determine whether any two mutations have occurred in the same cell.

Another promising avenue involves leveraging multiple tissues from the same individual. In the SMaHT Network, up to 15 tissues per donor are sequenced when available. Early developmental mutations are expected to be shared across multiple tissues, while tissue-specific or germ layer-specific variants are likely confined to a subset. This tissue distribution can be exploited to improve variant filtering, for example, by using one tissue as a control for another in paired-calling frameworks to distinguish true tissue-specific variants from technical artifacts.

Limitations of the study

While well-suited for controlled mixing and orthogonal validation, the COLO829BLT50 (1:49) and HapMap six-line mixtures contain artificially constructed mosaicism and do not recapitulate the spatial clonal architecture, mutational signatures, or chromatin context of *in vivo* somatic mosaicism. Moreover, with these cell lines, it was possible to generate high-coverage PacBio HiFi data, which requires high-molecular-weight DNA. Obtaining such high-molecular-weight DNA will be challenging for certain tissues (e.g., fibrous tissues), making it difficult to generate a comprehensive set of tissues with long-read data. Although we maximized discovery sensitivity with four complementary Illumina callers, there may be variants missed by all four; therefore, reported sensitivity values should be interpreted within the boundaries of this reference set.

In summary, our study highlights the current limitations of using sequencing depth alone to detect low-VAF mosaic mutations using bulk short-read sequencing data and underscores the need for integrative approaches—combining ultra-deep sequencing, long-read validation, multi-tissue analysis, and advanced computational methods. The pure cell lines, cell-line mixtures, and curated reference sets developed here are publicly available and will serve as valuable resources for the community to benchmark and advance mosaic and low-VAF variant detection tools.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Peter J. Park (peter_park@hms.harvard.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

The whole-genome sequencing data generated by the SMaHT network (COLO829, COLO829BL, COLO829BLT50, and HapMap mixture) can be found at https://data.smaht.org/. The reference sets for COLO829BLT50 and HapMap mixture can be found at GitHub (https://github.com/parklab/SMaHT_SNV_COLO829BLT50_HAPMAP) along with negative controls and cell-culture-derived mutations for COLO829BLT50. The reference genome GRCh38 (GenBank accession: GCA_000001405.15) is available at https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/. Different genomic categorizations of easy, difficult, and extreme regions can be found at https://github.com/parklab/SMaHT_Regional_Categorization. All pipelines and in-house scripts used for analysis can be found at https://github.com/parklab/SMaHT_SNV_COLO829BLT50_HAPMAP and are archived at Zenodo (https://zenodo.org/records/22113774).

ACKNOWLEDGMENTS

This research is supported by the NIH Common Fund, through the Office of Strategic Coordination/Office of the NIH Director under awards 1UM1DA058230, UM1DA058220, 1UG3NS132134, 1UM1DA058229, 1U24NS132103, UM1DA058235, UM1DA058236, UM1DA058237, and UM1DA058219. Y-J.J.H. was also supported by research funding from Hanyang University (HY-202500000003359) and the Bio & Medical Technology Development Program of the National Research Foundation (NRF), South Korea (RS-2024–00411768).

AUTHOR CONTRIBUTIONS

P.J.P., Y.-J.J.H., J.T.B., G.T.M., R.G., S. Germer, R.S.F., and K.A. supervised the project. Y.-J.J.H. and D.M. generated the reference set and conducted all analysis for the COLO829BLT50 benchmark. J.M. performed analysis on HapMap mixtures with reference set generation. S.J.G. carried out cell-culture-derived mutation analysis. N.L.P., T.H.H.C., S.D., S. Gardiner, D.K., D.L., B.M., R.M., L.X., Z.Y., G.N., and K.W. participated in the ultra-low-VAF mosaic mutation calling challenge. L.A., N.Y.T.A., C.C., H.D., C.M.G., D.M.J., K.L., A.N., J.O., M.M.P, A.M.R., A.B.S., L.M.S., and T.W. contributed to biological material and data generation. H.J. performed signature analysis on COLO829BLT50. Y.Z. developed a pileup script for indel analysis for short- and long-read data. T.L. assisted with consortium organization. C.T.K. contributed to regional stratification with Y-J.J.H. M.B., W.C.F., and A.D.V. contributed to data processing at the SMaHT Data Analysis Center under the supervision of H.-J.E.C. and P.J.P. Y.-J.J.H. and P.J.P. drafted the manuscript with contributions from D.M., J.M., S.J.G., and G.T.M.

DECLARATION OF INTERESTS

P.J.P. is a member of the Scientific Advisory Board for Bioskryb Genomics. J.T.B. is a consultant for Mosaica Medicines. A.B.S. is a co-inventor on a patent relating to the Fiber-seq and DAF-seq methods. D.K.'s immediate family member is employed at Oxford Nanopore Technologies.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.xgen.2026.101390.

Received: October 20, 2025
Revised: April 12, 2026
Accepted: September 12, 2026

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Please cite this article in press as: Ha et al., Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures, Cell Genomics (2026), https://doi.org/10.1016/j.xgen.2026.101390

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
COLO829 (melanoma)	ATCC	Cat. # CRL-1974, Lot # 70024393, RRID:CVCL_1137
COLO829BL (lymphoblastoid)	ATCC	Cat. # CRL-1980, Lot # 70022927, RRID:CVCL_1999
COLO829BLT50 (49:1 mixture of COLO829BL and COLO829)	SMaHT GCC UW-SCRI	N/A
HG00438	Coriell Institute	Cat# HG00438, RRID:CVCL_P889
HG002	Coriell Institute	Cat# GM24385, RRID:CVCL_1C78
HG02257	Coriell Institute	Cat# HG02257, RRID:CVCL_Q703
HG02486	Coriell Institute	Cat# HG02486
HG02622	Coriell Institute	Cat# HG02622, RRID:CVCL_GV49
HG005	Coriell Institute	Cat# GM24631, RRID:CVCL_1C97
HapMap Mixture	SMaHT GCC WashU-VAI	N/A
Biological samples		
ST001-1A (Liver)	TPC/NDRI	N/A
Deposited data		
Data for COLO829, COLO829BL, COLO829BLT50	dbGaP	phs004193
Data for COLO829, COLO829BL, COLO829BLT50	SMaHT DAC	data.smaht.org
Data for HapMap Mixture	dbGaP	phs004193
Data for HapMap Mixture	SMaHT DAC	data.smaht.org
Software and algorithms		
BWA-MEM (v0.7.17)	Li and Durbin[26]	https://github.com/lh3/bwa
pbmm2 (v1.13.0)	PacBio	https://github.com/PacificBiosciences/pbmm2
TNHaplotyper2 (Sentieon Genomics 202112.06)	Freed et al.[27]	https://support.sentieon.com/docs/usages/driver/TNhaplotyper2/
Strelka2 (v2.9.2)	Kim et al.[28]	https://github.com/Illumina/strelka
VarNet (v1.1.0)	Krishnamachari et al.[29]	https://github.com/skandlab/VarNet
RUFUS (v0.1)	Andrew Farrell	https://github.com/jandrewrfarrell/RUFUS
Manta (v1.6.0)	Chen et al.[30]	https://github.com/Illumina/manta
Variant validation using long reads	This paper	https://github.com/parklab/SMaHT_SNV_COLO829BLT50_HAPMAP
Haplotyper (Sentieon Genomics 202112.06)	Freed et al.[27]	https://support.sentieon.com/docs/usages/driver/Haplotyper/
Mutect2 (v4.4.0.0), (v4.1.6.0)	Benjamin et al.[16]	https://gatk.broadinstitute.org/hc/en-us/articles/13832655155099–Tool-Documentation-Index
MuSiCal	Jin et al.[31]	https://github.com/parklab/MuSiCal
Dynamic Read Analysis for GENomics (DRAGEN) (v4.3)	Illumina	https://www.illumina.com/products/by-type/informatics-products/dragen-secondary-analysis.html
Lancet2 (2.8.5-main-cfbc7d5143)	Musunuri et al.[32]	https://github.com/nygenome/Lancet2
MosaicForecast	Dou et al.[15]	https://hub.docker.com/r/yanmei/mosaicforecast
TNFilter (Sentieon Genomics 202308.03)	Freed et al.[27]	https://support.sentieon.com/docs/usages/driver/TNfilter/
ClairS-TO	Chen et al.[33]	https://github.com/HKU-BAL/ClairS-TO
Code for data analysis and variant calling	This paper	https://github.com/parklab/SMaHT_SNV_COLO829BLT50_HAPMAP
Code for region categorization	This paper	https://github.com/parklab/SMaHT_Regional_Categorization

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EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Experimental design

The SMaHT consortium designed several benchmarking experiments[34] to evaluate somatic variant detection. Here, we present the analyses of the two cell-line mixes based on (1) COLO829 melanoma and normal cell lines, and (2) selected HapMap cell lines.

The COLO829-based benchmarking experiment is a biologically controlled cell line mixture composed of 2% COLO829 (melanoma) and 98% COLO829BL (matched normal) and is referred to as COLO829BLT50. This design preserved the original germline haplotypes while introducing COLO829-specific mutations at 1% or 2%, corresponding to heterozygous and homozygous COLO829-specific mutations, respectively.

The HapMap mixture benchmarking experiment is a mixture of six clonal, well-characterized HapMap cell lines (HG002, HG005, HG00438, HG02257, HG02486, and HG02622) combined at different proportions to create artificial somatic variants spanning VAF ranges below 1%, 1–2%, 5–10%, and above 10%. This mixture was intended to establish lower limits of detectability for sequencing technologies and somatic variant callers. Together, the heterozygous and homozygous germline variants, both private and recurrent across these cell lines (Table S8), resulted in artificial somatic variants spanning VAFs from 0.25% to 16.5% (Figures S15A and S15B).

Cell line acquisition and mixture preparation

Generation of the COLO829BLT50 mixture

COLO829 (human male cutaneous melanoma cell line, Cat. # CRL-1974, Lot # 70024393) and COLO829BL (matched normal lymphoblastoid cell line from the same male donor, Cat. # CRL-1980, Lot # 70022927) cell lines were obtained from ATCC, expanded separately, and cryopreserved. For generation of the BLT50 mixture, cryopreserved cells were thawed, and viable cells were quantified using trypan blue staining and automated cell counting. Cells were mixed at a 1:49 ratio (COLO829:COLO829BL) based on viable cell counts, yielding a final tumor fraction of approximately 2%, and aliquoted into cryovials under constant agitation to maintain homogeneity.

Generation of the HapMap mixture

Human lymphoblastoid cell lines HG005 (male, Cat# GM24631, RRID:CVCL_1C97), HG02622 (female, Cat# HG02622, RRID:CVCL_GV49), HG002 (male, Cat# GM24385, RRID:CVCL_1C78), HG02257 (female, Cat# HG02257, RRID:CVCL_Q703), HG02486 (male, Cat# HG02486), and HG00438 (female, Cat# HG00438, RRID:CVCL_P889) were acquired from Coriell Institute, and cultures were separately thawed, recovered, and expanded. Cell lines were separately cultured and viable cell concentrations were quantified using a Vi-Cell Blu cell counter (Beckman Coulter) with trypan blue staining. Cells were pooled by cell line for cell counting and sterility assessment. Cell counts informed the necessary volumes of each cell line culture to generate the mixture at target mix ratio: 83.5% HG005, 10% HG02622, 2% each of HG002, HG02257, and HG02486, and 0.5% HG00438. The cell line mixture was centrifuged, resuspended in cryopreservation medium, and dispensed into cryovials at 5×10^6 cells/vial. Cryopreserved aliquots were stored in liquid nitrogen vapor.

Benchmarking donor tissue ST001-1A

The SMaHT Benchmarking donor tissue sample ST001-1A was obtained *postmortem* from the liver of a healthy 22-year-old male donor. PacBio long-read data generated from a homogenized sample of this tissue was downloaded from the SMaHT Data Portal (https://data.smaht.org/) to be used as control in this study.

METHOD DETAILS

COLO829BLT50 karyotype characterization

Cytogenetic analysis confirmed that COLO829 exhibits complex aneuploidy with both numerical and structural abnormalities, whereas COLO829BL is predominantly diploid, with a minor tetraploid subpopulation (Figure S1). To validate the effective tumor fraction and expected VAFs in the BLT50 mixture, droplet digital PCR (ddPCR) was performed for five known COLO829-specific mutations located in both euploid and aneuploid regions (Figure S9). The observed VAFs in the mixture were consistent with expectations for a 1:49 cell-level mixture within experimental variance, supporting the use of the nominal mixing ratio for benchmark construction.

Karyotype considerations and impact on expected VAFs in cell-line mixture

Cytogenetic analysis confirmed that COLO829 is highly aneuploid, exhibiting extensive numerical and structural chromosomal abnormalities, whereas COLO829BL is predominantly diploid (46, XY), with a minor tetraploid subpopulation detected in a small fraction of cells (Figure S1).

The COLO829BLT50 mixture was generated at the cell level, not by DNA mass. Cryopreserved COLO829 and COLO829BL cells were thawed, viable cells were quantified using trypan blue staining, and cells were mixed at a 1:49 ratio based on viable cell counts. Because COLO829 is aneuploid, local copy-number changes and LOH could, in principle, shift the locus-specific expected VAF of tumor-specific variants away from diploid expectations. We therefore evaluated whether COLO829 copy-number variation could materially affect the expected VAFs of COLO829-specific mutations in the COLO829BLT50 mixture.

Please cite this article in press as: Ha et al., Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures, Cell Genomics (2026), https://doi.org/10.1016/j.xgen.2026.101390

Given the low tumor fraction (~2% COLO829 cells mixed 1:49 with COLO829BL), any copy-number–driven shifts are expected to be modest in absolute terms relative to the ultra-low VAF regime used for benchmarking. To empirically assess this, we performed droplet digital PCR (ddPCR) for five known COLO829 mutations located in both euploid and aneuploid genomic regions. As summarized in Table S4, the observed VAFs in the COLO829BLT50 mixture closely matched the expected values for a 1:49 cell-level mixture, with deviations within experimental variance, regardless of underlying copy-number status. Together, these results indicate that, at the low tumor fraction used in this benchmark, aneuploidy in COLO829 does not substantially distort the designated low-VAF ranges.

For clarity to end users, we emphasize that downstream analyses in this study rely on empirically observed VAFs from high-coverage sequencing, rather than theoretical VAFs derived from the nominal mixture fraction under a diploid assumption. Users who require an "expected VAF" for modeling should incorporate local copy-number state (or restrict to near-diploid regions), as locus-specific deviations can occur in copy-number altered or LOH segments even when the overall mixture fraction is fixed.

DNA sequencing and uniform data processing

Aliquots of COLO829, COLO829BL, COLO829BLT50, and HapMap cell line mixture were distributed to various SMaHT Genome Characterization Centers (GCCs) for Illumina short-read and PacBio long-read whole-genome sequencing (WGS) at different target depths (see Table S1; Figure S2). Sequencing, data processing, and alignment for all samples analyzed in this study are described in the SMaHT Network paper[34] in detail. In brief, all WGS data were uniformly processed using standard pipelines for each sequencing platform: Illumina paired-end WGS data were processed using a pipeline based on the Genome Analysis Toolkit (GATK) Best Practices.[35,36] After raw sequence quality assessment, sequence artifacts are removed using fastp.[37] Reads were then aligned to the GRCh38 reference genome that excludes ALT contigs and decoy sequences using BWA-MEM[26] (*v0.7.17*), followed by uniform post-processing steps for data quality assessment, including duplicate reads marked using Picard,[38] local realignment, and base alignment quality and base quality score recalibration (BQSR). PacBio HiFi WGS data were aligned to the same reference genome using pbmm2[39] (*v1.13.0*).

WGS data acquisition

All Illumina short-read and PacBio long-read WGS data analyzed in this study can be obtained from the SMaHT Data Portal (https://data.smaht.org/). All libraries from COLO829 and the HapMap mixture were used. For the COLO829BL Illumina WGS data specifically, three BAM files, each derived from a sequencing library from the same COLO829BL analyte, were analyzed (file accession IDs: SMAFIARM7511, SMAFIJ9GV6FG, and SMAFIH6KN5FI). All COLO829BL PacBio WGS data was analyzed.

Quality controls for sequencing data

All WGS data were evaluated for data quality and sample integrity checks (e.g., sample swap or duplicates, contamination checks) using standard tools such as FastQC,[40] Samtools,[41] and NanoPlot,[42] as well as in-house scripts implemented at the SMaHT Data Analysis Center. For PacBio and ONT data, reads were evaluated to assess the read-length distribution, N50, and other QC metrics specific to long-read data. For Illumina data generated on the NovaSeq platforms, raw reads were evaluated for base quality and base composition and reads containing poly-G artifacts were removed using FastQC[40] (v.12.0) and fastp, respectively. The full description of the data quality control (QC) and assessment at the SMaHT Data Analysis Center is described in the SMaHT Benchmark Flagship paper.[34]

COLO829BLT50 reference set generation

Illumina short-read-based candidate selection

To identify somatic SNVs and indels unique to COLO829, we applied four somatic variant callers, TNHaplotyper2[27] (*Sentieon Genomics 202112.06*), Strelka2[28] (*v2.9.2*), VarNet[29] (*v1.1.0*), and RUFUS[43] (*v0.1*), using COLO829BL as the matched control. Each caller was executed with its recommended settings with 30Mb intervals and accompanied by internal variant filtering. For Strelka2, small indels were obtained using Manta[30] (*v1.6.0*). To ensure consistent representation of variants across tools, all outputs were normalized with bcftools[44] (*v1.14*; *norm -a -m - -f*). Only *PASS* calls were used, and Indels were restricted to those <50 bp in length.

Variant support validation with PacBio long reads

Variant support from PacBio HiFi data was assessed using bcftools mpileup[44] *(v1.17)*, applying filters of maximum depth 3,000, minimum base quality 20, minimum mapping quality 1, and excluding supplementary, secondary, duplicate, and QC-failed reads. Resulting pileups were normalized with bcftools *norm* (allele splitting and left alignment) to generate per-chromosome PacBio pileup VCFs for both COLO829 and COLO829BL. For SNVs, we required detectable alternative-allele support in the PacBio tumor pileup (alt read count >1), absence of alternative-allele evidence in matched normal blood (no ALT reads across all competing alleles), and excluded sites where the PacBio tumor pileup showed a different allele as the major alternative (multi-allelic mismatch).

PacBio HiFi (CCS) reads have a per-base error rate on the order of $\sim 10^{-3}$. At ~500×, the probability of observing at least one sequencing error at a given genomic position is substantial: $P(\geq 1 \text{ error}) = 1-(1-10-3)500 \approx 0.39$. Because our orthogonal validation

requires an error that matches the exact alternate allele, we further considered a simple uniform substitution model in which the probability of a given spurious alternate is $p \approx 10–3/3$ per read, resulting in ≈ 0.15. Requiring ≥ 2 independent supporting reads reduces this probability to P($\geq$2 error) = 1−P(0 error)-P(1 error) = ~1% and P($\geq$3 error) to $\sim 7 \times 10^{-4}$. Therefore, we required a minimum of two supporting PacBio reads for validation.

For indels, we developed a dedicated long-read–aware annotation framework designed to recover strand-consistent, breakpoint-resolved indel evidence from PacBio HiFi and Illumina short-read data. A custom Python module (Python v3.9.7, pysam[45] v0.21.0) processes each indel record by retrieving all reads overlapping the predicted breakpoint and examining their alignment patterns in the corresponding BAM files. For each indel candidate, allele-specific evidence was evaluated across PacBio HiFi tumor reads, Illumina tumor reads, and matched Illumina blood reads. At each candidate site, all primary alignments with MAPQ ≥ 1 and passing standard quality filters were extracted from the corresponding BAM files. Reads were classified as supporting the reference allele, the targeted alternative allele, or any non-target alternative sequence based on alignment at the indel breakpoint. For each dataset, we calculated total depth, reference-supporting reads, alternative-supporting reads, non-target alternative counts, and the resulting VAFs. High-confidence somatic indels were defined as those with clear support in both long-read and short-read tumor data (PacBio VAF ≥ 0.20, Illumina tumor VAF ≥ 0.20) while showing minimal signal in matched normal blood (Illumina blood VAF <0.05). Indels failing the tumor-support thresholds or exhibiting alternative-allele signal in blood were excluded as non-somatic events.

COLO829BLT50 negative control construction

Nonvariant sites

To define the negative control set, we first identified homozygous reference sites shared by COLO829 and COLO829BL Illumina WGS. To define potential non-variant positions shared by COLO829 and COLO829BL, Sentieon Haplotyper[27] in the GVCF mode was applied independently on the two samples and putative homozygous positions shared by both were intersected for further processing. Next, we applied *bcftools mpileup* on the alignment of COLO829 and COLO829BL PacBio WGS in the putative homozygous candidates with the minimum mapping quality for an alignment (*q* parameter) and minimum base quality for a base to be considered (*Q* parameter) set to 1. Positions where no alternate allele was found in both COLO829 and COLO829BL were collected as homozygous reference sites and included in the negative control set. The rest of the positions were then subjected to the next iteration of the *mpileup* analysis but with increased *q* and *Q* parameters (both set to 30). If alternate allele was observed with the higher quality threshold, those were removed from the negative control. Then, we also checked Illumina WGS data to confirm remaining ambiguity. We again utilized *bcftools mpileup* on the short-reads with adjusted *q* and *Q* parameters (set to 1) and juxtaposed with the results of PacBio WGS pileups. Variants supported by both platforms were excluded, whereas those lacking Illumina support were retained as high-confidence negatives.

Germline variants

We collected germline variants in COLO829BL with Sentieon Haplotyper.[27] Filtration involved several steps: First, GATK (*v4.1.9.0*) hard filters were applied based on GATK best practice recommendations (*QD* < 2.0, *FS* > 60, *DP* < 10, *ReadPosRankSum* < −8.0, and *MappingQualityRankSum* < −2.5 or >2.5), retaining only PASS variants. Second, to remove germline-like systematic errors specific to PacBio data, particularly in homopolymer and tandem repeat regions, a binomial test [Binom(N, k, p) = Binom(depth, alternative allele count, 0.5)] with a *p*-value threshold of 0.05 was applied to distinguish heterozygous indels from subclonal Indels and artifacts.

Characterization of COLO829BLT50 artifacts

Generating admixture-only variant sets

Artifacts of the COLO829BLT50 mixture were first suspected after comparing the RUFUS-reported VAF from variants generated from the *in-silico* COLO829BLT50 datasets with the physical admixture datasets. Within only the COLO829BLT50 admixture datasets, we observed variants reporting allele fractions in the 5–10% range, i.e., allele frequency values seemingly inconsistent with the 1:49 tumor/normal blood cell line mixture. Manual inspection of several sites revealed the absence of the called mutant allele from the tumor sequencing dataset. To further investigate these variants, we examined call sets generated by both RUFUS and Mutect2 (performed in COLO829BLT50/BL paired calling mode) within each GCC-specific replicate COLO829BLT50 datasets. RUFUS and Mutect2 call sets within each replicate, were normalized and merged, creating a distinct call set within each of the five replicates. Any variants within these call sets that existed within the COLO829/COLO829BL reference set were removed, leaving only COLO829BLT50 admixture-specific variants. We then created 5-way intersections of these replicate call sets, as shown in the upset plot in Figure 3C. We identified sites present in all five replicate datasets, i.e., C(5,5); then those present in 4, 3, 2, or only 1 of the 5 replicate datasets, i.e., C(5,4), C(5,3), C(5,2), and C(5,1), with the hypothesis that sites called within fewer replicate datasets represented lower confidence for the existence of the variant in the COLO829BLT50 mix.

Profiling admixture variants in PacBio

To characterize each site, we performed a pileup in a 390× deep PacBio COLO829BLT50 dataset generated with the PacBio Revio sequencing instrument, and tabulated the number of PacBio reads supporting the mutant allele at the same site. We identified sites for which at least two supporting long reads were present. To avoid spurious confirmation caused by genomic site-specific sequencing errors, we repeated this procedure in a control sequencing dataset from a human liver tissue homogenate sample from an unrelated

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individual (ST001-1A) 200× depth, which was prepared using identical DNA extraction, library construction, and sequencing protocols to the COLO829BLT50 mix. We identified all sites at which the mutant allele called in the COLO829BLT50 dataset was supported by at least two PacBio long reads in the negative control dataset. For indel confirmation, we additionally required that the mutant allele be the most frequently observed allele at the site, or second in frequency only to the reference allele, to guard against alignment artifacts that can produce spurious indel support in repetitive or low-complexity regions. We further cross-validated indel candidates against Illumina short-read tumor sequencing data (~200×), requiring a minimum site depth of 10 reads and at least two reads supporting the mutant allele, and excluded candidates with reference or alternative allele lengths exceeding 50 bp.

Defining the admixture-only reference set

We have finally identified all COLO829BLT50 admixture-only candidate variants with a minimum of two supporting long reads in the COLO829BLT50 dataset, excluding sites with spurious "confirmation" in a control dataset. We have carefully identified the provenance of each reference-set variant whether it originated in the COLO829/COLO829BL-generated reference set or in the COLO829BLT50 admixture-only reference set. The identified sites were examined using both PacBio and Illumina alignments of COLO829 and COLO829BL. The datasets were queried using *bcftools mpileup* with filters of minimum mapping quality of 1 (*--min-MQ* 1) and minimum base quality of 20 (*--min-BQ* 20). Variants present in COLO829BL but absent in COLO829, with at least 1 read of support in the sequencing dataset, were categorized as COLO829BL-only mutations. Mutations with no evidence in either COLO829 or COLO829BL were intersected with the negative control regions to retain only high confidence variants forming the COLO829BLT50 admixture-only mutation reference set.

Mutational signature analysis

Mutational signature analysis was performed on false positive (FP) mutation calls annotated as "nonvariant". A mutation count matrix was constructed by stratifying these FPs by sample, VAF bin, and variant calling pipeline. *De novo* mutational signatures were extracted from this mutation count matrix with MuSiCal.[31] Two of the discovered signatures – showing similarities to COSMIC SBS18 and SBS17b – were likely associated with ROS damages in cell cultures and shown in Figure S5.

In silico COLO829BLT50 generation

The COLO829 and COLO829BL Illumina data used for reference set generation were downsampled with picard DownSampleSam (STRATEGY = HighAccuracy) with 2% and 98% respectively to the target depths: 100×, 200×, 300×, 400×, and 500×. The depths of the downsampled BAM files were measured with picard CollectWgsMetrics to confirm the accuracy of downsampling. Then, those BAM files were merged with samtools merge. For downsampling and merging, 30 Mb intervals were used.

Genomic categorization of easy, difficult, and extreme regions

Definition of three genomic regions

To evaluate low-VAF mosaic SNV and Indel detection performance across genomic contexts, we stratified the genome into three categories reflecting practical variant callability: easy, difficult, and extreme regions. Easy regions correspond to genomic intervals defined by the 1000 Genomes Project strict accessibility mask, representing confidently mappable and consistently variant-callable loci across population samples. Difficult regions include loci present in the PanMask pm151 (strict) set but excluded from the 1000 Genomes strict mask. These regions are moderately mappable and often exhibit elevated error rates in short-read variant calling. Extreme regions comprise the remaining genome outside both masks and include highly repetitive or structurally complex loci that remain poorly resolved, precluding accurate low-VAF variant detection.

Relationship to GIAB stratifications

The GIAB/GA4GH genomic stratifications (e.g., Dwarshuis et al.[46]) provide widely adopted annotations for analytical validation, including labels based on sequence and mapping features such as GC content, mappability, and low complexity/repeat content. These are commonly used in clinical short-read variant calling to identify regions with elevated technical artifacts and reduced precision. Our stratification is complementary in scope: rather than enumerating individual features, we aggregate them into three benchmark-oriented callability tiers tailored to ultra-low-VAF detection, where modest differences in mappability and alignment ambiguity can dominate performance.

Rationale for pm151

The operational definition of "difficult" regions evolved during this study. We ultimately adopted pm151, because it is informed by mapping/callability across ~500 diverse genomes, providing a robust empirical delineation of regions where 151-bp reads can be mapped and variants can be called with relatively high accuracy across many samples. In this sense, the resulting difficult/extreme partition provides a pragmatic improvement over earlier coarse classifications for short-read benchmarking in challenging regions. A detailed source mask and implementation details are available in the accompanying GitHub repository (link).

Mosaic mutation detection challenge pipelines

BCM (M1). The Sentieon-generated GRCh38 BAMs for COLO829BLT50 sample-mix, were merged from each of the five sequencing centers. Variant calling was performed using Illumina's Dynamic Read Analysis for GENomics (DRAGEN)[47] software *v4.3*. Mosaic detection was enabled (*–vc-enable-mosaic-detection true*) with defaults used for other variant calling options. Variants on autosomes and allosomes tagged as PASS, MOSAIC, with $AF \leq 0.05$ were retained using bcftools[44] *v1.19;* any multi-allelic variant calls were excluded. *In-silico* datasets were not processed at BCM-HGSC.

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Broad (M2, M3, and M4). Variant calling was performed using Illumina's Dynamic Read Analysis for GENomics (DRAGEN) software *v4.3* for all cell-admixture and *in-silico* COLO829BLT50 samples. To ensure all reads were assigned to a single read group in BAM files of each sample, the *–use-single-read-group-for-bam-list* parameter was applied. Variants with VAF lower than 20% and tagged as PASS were retained (**M2** in Table 1). The same pipeline was applied on a single COLO829BLT50 sample sequenced at the Broad Institute but without the prepreprocessing of the read group information in BAM files (**M3** in Table 1). For M4, variant calling was performed with GATK Mutect2 on the Terra platform (v4.1.6.0; github.com/broadinstitute/gatk/mutect2). To filter likely germline variants from the putative somatic calls, a 1000 Genomes panel of normals and the Genome Aggregation Database (gnomAD) were utilized. Mutect2 calls were subsequently subjected to filtering and refined to retain high-confidence variants based on the following strict criteria: 1) exclusion of regions with very low (<50×) or abnormally high (>1000×) sequencing coverage; and 2) removal of residual germline variants using a one-sided exact binomial test assuming an expected germline VAF of ≥0.5, followed by a Benjamini-Hochberg correction with a q-value cutoff of 10^{-5}.[19]

NYGC (M5). Mosaic SNVs and Indels were detected using Lancet2[32] (*2.8.5-main-cfbc7d5143*) in the single sample mode. The candidates were subjected to a series of following filters – Germline variants, identified with HaplotypeCaller and GenotypeGVCFs, were filtered out. Variants present in either gnomAD v3.1.2 (with allele frequency greater than 1*10–5%) or in the high coverage NYGC generated 1000 Genomes call sets were removed. Variants in simple repeat (UCSC) and segmental duplication regions were excluded. Candidate sites were retained if they had a total depth between 0.65× and 1.25× of the mean sample coverage, with at least four supporting reads, and a variant allele fraction (VAF) ≤ 10%. Additional filters accounting for strand bias, base quality, and mapping quality were applied to retain only high-confidence mosaic SNV and Indel calls.

WashU-VAI (M6 and **M7).** Somatic SNPs and indels were identified from COLO829BLT50 BAM files using GATK Mutect2 (*v4.4.0.0*) in tumor-only mode, with the GRCh38 reference genome, incorporating the gnomAD germline resource, and a 1000 Genomes panel of normal. To improve computational efficiency, the genome was divided into 30 intervals with *SplitIntervals*, and Mutect2 was run in parallel. Interval-level variant calls and statistics were merged using *MergeVcfs* and *MergeMutectStats*, and high-confidence variants were obtained with *FilterMutectCalls*. Two complementary post-calling strategies were applied. In the first, variants with allele frequency <0.02 were retained using bcftools, consistent with the expected 1:49 tumor–normal mixture (**M6** in Table 1). In the second, variants were annotated against dbSNP151 using ANNOVAR,[48] and only novel SNPs and indels were retained (**M7** in Table 1). This set was further filtered to exclude calls with sequencing depth <50× or >1000×, allele frequency >20%, and variants outside autosomes, chrX, chrY, and chrM. The resulting VCFs contained the final set of high-confidence somatic variants used for downstream analyses.

UW-SCRI (M8). BAM files from the sequenced COLO829BLT50 resource from each GCC were analyzed with GATK Mutect2 in tumor only mode using a panel of normals and the gnomAD resource both downloaded from the Broad Institute to filter likely germline variants. Only *FILTER=PASS* variants were included using standard Mutect2 filters. The final call sets were filtered to require a depth >20 and VAF less than 40%.

Utah-TTD (M9, M10, and M11). COLO829BLT50 BAM files from a single GCC were initially merged for a single subject file input. RUFUS[43] was run in a parallelized 1Mb windowed mode, using COLO829BL as the paired control. Standard input parameters of a minimum k-mer threshold of 5 and k-mer length of 25bp were used, along with the *ReportLowFrequency* flag to allow identification of somatic mutations. After all 1Mb region runs were completed, the variant call sets were trimmed to exclude any reported variants outside of the specified region and then concatenated into a single genome-wide VCF. A standardized RUFUS post-process script was applied (included in the tool Singularity container) and indels and SVs were removed to result in a final SNV-only, genome-wide call set. RUFUS final VCF calls were evaluated using a variant quality score (VQS) classifier to remove potential false positive calls. The VQS model was a logistic regression trained on features from validated SNVs and indels from the SEQC2 HCC1395 tumor/normal benchmark dataset.[49] Three filtering stringencies were applied to the RUFUS call set. For the *lenient* threshold, no VQS filtering was applied (**M9** in Table 1). The *moderate* threshold corresponded to a VQS cutoff that achieved ∼90% sensitivity to true mutations for both SNVs and indels in the training data (**M10** in Table 1), while the *strict* threshold corresponded to cutoffs yielding ∼80% sensitivity for SNVs and ∼60% sensitivity for indels (**M11** in Table 1). These thresholds were designed to progressively reduce false positive calls while maintaining defined levels of sensitivity in mutation detection.

DAC (M12). MosaicForecast[15] was run within the official MosaicForecast Docker container (https://hub.docker.com/r/yanmei/mosaicforecast). We first ran Sentieon TNHaplotyper2[27] *v202308.03* (GATK Mutect2 equivalent) followed by Sentieon TNFilter *v202308.03*. To split multi-nucleotide and multi-allelic variants into separate records, output VCFs were processed with bcftools[44] (*v1.14*; *norm -a -m - -f*). Segmental duplication and variant clustered regions were removed from the VCFs before running MosaicForecast. The MosaicForecast *ReadLevel_Features_extraction.py* script was applied with the following parameters: (1) Umap mappability scores for k = 24 generated based on the repository's provided script, (2) candidate variants pre-formatted from VCF to BED, (3) the corresponding BAM file and (4) the reference genome. *Refine* models were selected based on the sequencing coverage of the input BAMs.

Evaluation of detection challenge pipelines

To harmonize variant representations, we used bcftools[44] (*v1.14*) to split multi-nucleotide variants (MNVs) and multi-allelic sites into standardized single-allele records. For all benchmarking analyses, we used the filtered subset of the reference set with expected VAF in the COLO829BLT50 mixture >0.002 (VAF_III × 0.02 > 0.002), excluding variants below the minimum non-zero VAF expected from a single alternate read at ∼500× coverage. Variant allele fractions (VAFs) were extracted directly from

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COLO829BLT50 BAM files using *bcftools mpileup* (base quality and mapping quality ≥1) for all mSNVs and mIndels in the reference set, including cell-culture-derived mutations. If reported mutations from pipelines were confirmed in the reference set, they were marked as true positives. If a mutation was found in negative controls, it was marked as false positive. If not found in both reference set and negative control, it was masked and not used for the evaluation due to ambiguity in the classification. For variants outside the reference set, VAFs were taken from the values reported by the respective callers. Sensitivity (true positives out of all reference set), precision (true positives divided by the sum of true and false positives), and F1 scores (balanced mean of sensitivity and precision) were computed across five VAF bins (0–0.5%, 0.5–1%, 1–1.5%, 1.5–2%, 2–3%), chosen to distribute reference set variants more evenly across bins and enable fairer comparisons among methods.

Evaluation across different genomic contexts

Easy regions were defined by the 1000 Genomes strict mask (high-confidence, short-read–callable regions). Difficult regions corresponded to sites within the PanMask[50] strict (pm151b) set but outside the 1KG strict mask, representing moderately mappable regions. PanMask, a recent pan-genome-based accessibility mask derived from hundreds of high-quality assemblies, was used to provide a more generalizable and less reference-biased framework for benchmarking. Extreme regions encompassed loci outside PanMask, typically highly repetitive or structurally complex. Detailed description and resources can be found on GitHub.[12]

Cross-sample artifact filtering workflow for short-read variant calls

To mitigate recurrent, alignment-driven false positive SNVs enriched in difficult and extreme genomic regions, we implemented a cross-sample pileup filtering workflow using an independent control sample. For candidate variants, we generated targeted pileups at variant loci using *bcftools mpileup* (v1.21; minimum base and mapping quality Q20, depth capped at 3,000, excluding secondary and duplicate reads). Pileups were normalized (*bcftools norm*) and restricted to candidate sites. Variants were retained only if supported by > 1 alternate read in the target sample and ≤1 alternate read in the independent control sample. This allele depth–based criterion suppresses recurrent low-level sequencing and alignment artifacts observed across datasets while preserving true positives. The workflow was implemented in Snakemake and is publicly available on GitHub.[20]

Analysis of HapMap mixture experiment

HapMap reference set generation

To generate a reference set of expected variants and their anticipated VAFs in the HapMap mixture, we first obtained germline variant call sets for the individual cell lines. Curated germline variants and sample-specific confident regions for HG002 and HG005 were obtained from the Genome in a Bottle (GIAB) benchmarking effort,[51] while those for HG00438, HG02257, HG02486, and HG02622 were obtained from the Human Pangenome Reference Consortium.[52] For all six cell lines, only SNVs within the intersection of the sample-specific confident regions were retained, covering 2.43 Gb of the genome. Indels, multiallelic variants, and SNVs with ambiguous reference bases (“N”) were excluded. Expected mixture VAFs were calculated based on the heterozygous or homozygous status of variants and the mixing proportions of the cell lines they occur in. In total, the HapMap mixture reference set contained 8,031,751 SNVs, with 4,888,479 within the artificial somatic VAF range (0.25–16.5%). Most of the somatic variants were private heterozygous variants from the 2% contributing cell lines, yielding >1,774,000 variants expected at 1% VAF.

Subsampling and merging of datasets

To assess the impact of sequencing depth on variant sampling and sensitivity of variant calling at low VAFs, short-read datasets were subsampled using *samtools view –subsample*[44](samtools *v1.17*) to 300× (WashU-VAI and BCM) and 100× (WashU-VAI, BCM, and NYGC). To evaluate whether aggregating lower-coverage datasets across GCCs could approximate higher-coverage sequencing by one GCC, all 100× subsampled datasets from different GCCs were combined to form a 300× aggregate dataset.

Variant allele sampling in HapMap mixture

To investigate the presence of expected variants in the HapMap sequencing datasets as a base for subsequent somatic variant calling, we examined allelic support at all HapMap reference positions in both Illumina and PacBio datasets. Pileups were generated using *bcftools mpileup*[44] (*v1.17*) with minimum base quality and mapping quality thresholds set to 30, ensuring only high-confidence bases in reliably aligned reads were considered.

Somatic variant calling in HapMap mixture

Many somatic variant callers were developed in a cancer research framework and are designed for tumor-normal comparisons. We applied callers with tumor-only functionality, given that the HapMap mixture mimics somatic variants without matched normals. Although the experimental design recapitulates low-VAF somatic variants, it differs from healthy human tissue samples in two aspects: the exceptionally large number of low-VAF variants (>4.8 million) and the presence of 12 haplotypes from six unrelated donors. To account for these design-specific aspects, we retained not only PASS variants but also those flagged for haplotype clustering (multiple nearby variants on the same haplotype) or variants annotated in germline datasets. Importantly, variants flagged with technical or quality-related filters (e.g., “base_qual,” “LowQual”) were excluded. Multiallelic variants were decomposed with *bcftools norm -m-any*, and indels as well as variants outside the confident regions were removed. Variants originating from HG005, the background cell-line, were considered germline and removed. False positives recurring across multiple samples were also excluded.

Please cite this article in press as: Ha et al., Detecting low-allele-fraction mosaic mutations: A multi-institution benchmarking with cell-line mixtures, Cell Genomics (2026), https://doi.org/10.1016/j.xgen.2026.101390

Illumina variant calling

Somatic variants in the Illumina datasets were called using *TNhaplotyper2* (Sentieon Genomics *202112.06*),[27] the Sentieon implementation of GATK Mutect2, in tumor-only mode. Variants flagged as "clustered_events" (indicating multiple events on the same haplotype) or "haplotype" (variants co-occurring with filtered variants on the same haplotype) were retained alongside PASS calls.

PacBio variant calling

PacBio datasets were analyzed with ClairS-TO[33] (Somatic Tumor-Only mode) using parameters –snv_min_af 0, –min_coverage 1, and –qual 4 to enable detection of variants at VAFs below 5%. Variants were not filtered against a panel of normals to avoid removing true positives. Calls flagged as "NonSomatic" (indicating overlap with germline SNV databases such as 1000 Genomes, gnomAD, or dbSNP) were retained, as the HapMap donor germline variants are expected to appear in these databases.

ADDITIONAL RESOURCES

All generated sequencing data from cell lines COLO829, COLO829BL, COLO829BLT50 mixture, in silico COLO829BLT50 mixture, and the HapMap mixture, as well as the reference sets for somatic variants calls for the COLO829BLT50 mixture and the HapMap mixture are provided on the SMaHT data portal at https://data.smaht.org/.